

Research Paper

Multi-objective optimization and posteriori multi-criteria decision making on an integrative solid oxide fuel cell cooling, heating and power system with semi-empirical model-driven co-simulation

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ABSTRACT

An integrative solid oxide fuel cell combined cooling, heating and power system in green buildings with hydrogen energy of byproduct water enables carbon neutrality transformation. However, underlying mechanisms on capacity sizing of combined cooling, heating and power system devices and its impacts on system techno-economy have not been figured out especially considering dynamic degradation and efficiency of associated devices. In this study, a multi-software optimization platform is established by MATLAB-TRNSYS co-simulation for sizing parametrical analysis, with well balance of modelling complexity and computational efficiency. A self-sufficient combined cooling, heating and power system is modelled integrating with a semi-empirical surrogate model of solid oxide fuel cell to interact with other balance of plant types efficiently. Total energy efficiency and annual total cost are optimized through parametrical analysis on device size of each component (battery, electrolyzer and solid oxide fuel cell) and analysis of variance for contribution ratio quantification. Results indicate that, the size increase in electrolyzer and solid oxide fuel cell will improve system total energy efficiency by 13.635 % and 2.194 %, but promote annual total cost by 4.042×10^4 \$ and 2.389×10^3 \$, respectively. Besides, sensitivity analysis indicates that the electrolyzer size prioritizes other design parameters in techno-economic performance. Optimal sizes of battery, electrolyzer and solid oxide fuel cell are in cell number range of 333 – 403, 17 – 20, and 26 – 30, respectively, with corresponding optimal total energy efficiency and annual total cost at 70.861 % – 72.147 % and 6.723×10^4 \$ – 7.325×10^4 \$, respectively. The research results can provide guidance on hydrogen-based cooling, heating and power system design and operation with techno-economic feasibility for low-carbon district energy transition.

1. Introduction

As a common pursuit of carbon neutrality target around the world [1], hydrogen energy has experienced substantial growth in both industrial sectors and academic circles in recent years [2], due to its prominent merits of extreme clean [3], high enthalpy [4], carbon free and easy long-term storage [5]. Solid oxide fuel cell (SOFC) has been attracted increasing attention because of its ability to convert hydrogen

into heat and electricity with high-quality heat. SOFC stack consists of series of cells containing interconnected bipolar plate, anode and cathode electrodes, and electrolyte for transporting ions. Wherein hydrogen will be oxidized and release electrons in anode. The released electrons will be transported into cathode by outside circuit, and reduce oxygen into ions in cathode. Afterwards, the oxygen ions will be transported to anode across the electrolyte, and combined with hydrogen ions to generate water. There are several types of SOFC, such as samarium-doped ceria (SDC) and yttria stabilized zirconia (YSZ) for electrolyte

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Nomenclature		Subscripts and superscripts	
A	Area, m^2	a	Anode
C	Specific heat capacity, $J/(kg \cdot K)$	act	Active
Ca	Battery capacity, kWh	am	Ambience
CN	Cycle number	ba	Battery
$Cost$	Cost, \$	c	Cathode
CR	Capital recovery factor	cell	Cell
D	Depth of discharge	co	Consumption
d	Thickness, m	com	Compressor
E	Potential; overpotential, V	con	Concentration
ΔE	Voltage degradation, V	cr	Constant rated power state
F	Faraday constant, C/mol	ct	Constant turning power state
f	Energy efficiency	de	Degradation
I	Current, A	dep	Depreciation
i	Current density, A/m^2	eff	Effective
L	Lifetime	el	Electrolyzer
LHV	Low heat value, J/kg	ele	Electrolyte
M	Molar mass, kg/mol	fl	Low power fluctuation state
m	Mass flow rate, kg/s ; mass, kg	fh	High power fluctuation state
mf	Mass fraction	ge	Generation
N	Cell number	HE	Heat exchanger
n	Electron number	in	Inlet
P	Power, W	idling	Idling state
p	Pressure, atm	ini	Initial state
Q	Cooling capacity, kW	int	Interconnector
R	Gas constant, $J/(mol \cdot K)$	lim	Limiting current density
RC	Relative capacity	loss	Loss
ΔRC	Degradation rate	main	Maintenance
r	Interest rate	max	Maximum
S	Surface area, m^2	mt	Maintenance state
T	Temperature, K	ner	Nernst
t	Time period, h	ohm	Ohmic
W	Energy, J	out	Outlet
Greek symbols		peak	Peak point
α	Stoichiometric ratio	ra	Radiation
β, γ, δ	Coefficient	re	Reversible potential
ζ	Maintenance coefficient	s	Solid oxide fuel cell
κ	Electron conductivity, S/m	sem	Semi-empirical
λ	Membrane water content	sys	System
μ	Degradation rate, $\mu V/h$	valley	Valley point
σ	Ionic conductivity, S/m	t	Thermal; time point
ω	Degradation rate, $\%/h$	total	Total

material. Compared to YSZ-based SOFC with operating temperature above 800 °C, SDC-based SOFC operates under the temperature of 500 °C–800 °C [6], which is much more time- and energy-saving during start up and long-time operation, and gained increasing concentrations by academia recently [7].

The SOFC has been applied in combined heating and power (CHP) system or combined cooling, heating and power (CCHP) system to cover the daily energy demands of buildings, contributing significantly to carbon neutrality transformation in green buildings [8]. The CCHP/CHP system will be integrated with other devices to adapt to variable scenarios, such as photovoltaic (PV) panel [9], battery [10] and electrolyzer [11], for long-time standalone operations. Those key device sizes, i.e., the cell number of the battery, electrolyzer, SOFC and etc., play a significant role in system economic feasibility and energy efficiency. Modeling analysis is an effective way to investigate the device size effects, with attributes of high efficiency and cost saving compared to in-field experiments [12]. By relying on computational simulation, it is necessary to guarantee the detailed information of integrated

components and calculation efficiency of the CCHP/CHP system modeling for overall performance analysis.

Besides, the capacity size of key devices will determine the investment cost of the CCHP/CHP system [13], and affect the system economic efficiency and implementation feasibility. On the other hand, the depth of discharge (DOD) and service lifetime (like battery and SOFC) will also be dependent on their sizes [14]. Therefore, it is necessary to design appropriate sizes of key devices to avoid performance overestimation and improve techno-economic feasibility.

Optimization analysis is an effective way to identify the optimal designs of CCHP/CHP systems, including operating control [15], material selection [16] and structure size [17]. High-dimensional modeling is usually used for device performance analysis and figuring out the system detailed characteristics by optimization. Pang et al. [18] optimized a steam ejector applied in a CHP system by comparing three design cases, through the finite element modeling and computational fluid dynamics analysis in Fluent software, which will consume a large amount of time for device performance calculation. They found that the

ejector designed with the optimized thermodynamic method possessed higher heat recovery efficiency. Gao et al. [19] proposed a cross-scale model of CHP system and established a fuel cell device model by computational fluid dynamics analysis in Fluent software, aiming to improve the fuel cell model accuracy. It is distinct that detailed modeling in high-dimensional is time-consuming for calculation because it needs to be solved by nonlinearly iterative calculation based on gradient descent method in each boundary condition solving, not to mention the global optimization by algorithm. While in order to achieve high-efficiency global optimization, the system devices are usually established in zero dimension, such as physical model or machine learning model. Yuan et al. [20] optimized a CHP system energy efficiency based on an artificial neural network model of input operating parameters and output efficiency relationship through the genetic algorithm calculation. The results showed that the system energy efficiency reached 95.62 %. Liu et al. [21] developed a zero-dimensional SOFC CCHP system model to analyze the thermal-economic performance by multi-objective optimization. They found that the total energy output and exergy efficiency could increase by 35.58 kW and 0.20 %, respectively. Besides, some research also explores methods to balance the calculation efficiency and model complexity. Darzi et al. [22] proposed a partial high-dimensional modeling method to optimize the spark plug location in CHP system, namely one-dimensional modeling of engine and three-dimensional modeling of cold flow in engine space, which can improve the calculation efficiency and guarantee the model accuracy to some extent, whereas the calculation efficiency is still limited. Obviously, high-dimensional model is hard to be applied in long-time scenario simulation and optimization, and device modeling is usually simplified to achieve a high-efficiency calculation in global optimization, with sacrifice or simplification of the device detailed characteristics. It is highly necessary to balance both model complexity and computational efficiency in the multi-scale optimization.

Sizing analyses of CCHP/CHP systems have been conducted to reduce total cost [23], improve energy saving [24], and reduce carbon emission [25], etc. Liu et al. [25] analyzed the capacity of prime mover in a CCHP system for optimal economic feasibility, and finally they identified the best capacity of the prime mover at 78.146 kW. Shahsavari et al. [26] analyzed the best energy cost of a CHP system in rural residential areas by sizing the system configuration, and they found that the optimal cost could be achieved through the design of 39.4-kW PV, 10-kW diesel generators, 20.1-kW converter, device number of 20 wind turbines and 96 batteries. Benalcázar et al. [27] proposed a mixed-integer linear programming approach to find the optimal capacity of a CHP system thermal energy storage. They found that the investment of thermal energy storage capacity would save the cost of € 0.57 M – € 0.64 M in Poland. Wang et al. [23] investigated the optimal size of equipment in the photovoltaic/thermal (PV/T) panel CCHP system by a bi-level optimization. The results indicate that the improvement of PV/T-CCHP system size could save the annual total cost of 5.82 % and reduce the carbon emission of 37.51 %. However, there are few studies about SOFC CCHP/CHP system sizing, and the CCHP/CHP system performance is highly dependent on the SOFC electrical/thermal efficiency, while the impacts of SOFC size on the SOFC electrical/thermal efficiency has been rarely discussed. Besides, the system sizing research has seldomly connected the device size design with its degradation performance, and this ignorance will lead to inaccurate estimations on the system lifetime and economic efficiency.

The collaborative sizing optimization of multiple devices in CCHP/CHP system is able to improve the system performance comprehensively [23]. Guo et al. [28] optimized the sizes of the proton exchange membrane fuel cell (PEMFC) and auxiliary chiller in a CCHP system. It indicated that the system energy efficiency could be improved by 23.21 %. Wang et al. [29] optimized the device capacity size of a CCHP system by multi-objective optimization algorithm for optimal carbon emission and economic cost. The results showed that the carbon emission and economic cost have been reduced by 99.81 % and 66.70 % respectively

compared to single objective optimization. Shahsavari et al. [26] identified the power capacity of PV panels, diesel generator, and converter, and installed number of wind turbines, battery and boiler in a CHP system. The results indicate that the energy cost could be reduced by 34.8 % under the optimal identification. Nonetheless, few studies have discussed the capacity sizing optimization of SOFC CCHP/CHP system. In addition, the contribution ranks of different key devices on CCHP/CHP system performances have not been figured out during the sizing optimization, especially combining the impacts of dynamic degradation condition and energy conversion efficiency.

Based on the above literature review, the research gaps can be summarized as follows:

- 1) The global optimization analysis of SOFC CCHP system has usually sacrificed or simplified the model complexity of detailed characteristics to guarantee the computational efficiency in zero-dimensional modeling, while the simplified model has limited the authenticity and reliability of the CCHP system model. It is necessary to well balance the model complexity and computational efficiency of the CCHP system.
- 2) In the academia, studies are quite limited on the sizing analysis regarding the integrative SOFC CCHP system, and the impact of device size (like battery, electrolyzer and SOFC) on their degradation performance has been usually ignored, resulting in inaccurate estimation of the overall techno-economic performance. Besides, the underlying mechanism of the SOFC size on the overall electrical/thermal efficiency has been seldomly investigated. Furthermore, sensitivity analysis of key device sizes on techno-economic performances of the integrative SOFC CCHP system has not been figured out.
- 3) In the academia, studies are quite limited on the size capacity optimization of integrative SOFC CCHP system, and the optimal designs of key device sizes to maximize techno-economic performances of the integrative SOFC-CCHP system have not been investigated, especially considering the dynamic degradation performance and energy efficiency simultaneously.

Main contributions of this study are summarized below:

- 1) In order to well balance both model complexity and computational efficiency of the integrative SOFC-CCHP system, a multi-software optimization platform is established based on MATLAB-TRNSYS co-simulation framework of a CCHP system coupling with a SOFC semi-empirical surrogate model. This aims to enable the multi-objective optimization with high computational efficiency and model complexity.
- 2) In order to clearly demonstrate the significance of key device size design (like battery, electrolyzer and SOFC) on the integrative SOFC-CCHP system, the impact of key device size analyses has been conducted on dynamic electrical/thermal efficiency and overall techno-economic performance of the integrative SOFC-CCHP system (like system energy efficiency, degradation level and annual cost), which also considers dynamic degradation performance. Furthermore, significance ranks of key device sizes on techno-economic performances of the integrative SOFC-CCHP system are identified through the analysis of variance (ANOVA) method, in terms of system total energy efficiency and annual total cost.
- 3) Multi-objective optimization and posteriori multi-criteria decision making on an integrative SOFC-CCHP system have been conducted. The optimal size designs are selected by normalized Euclidean distance along the multi-objective Pareto front to identify the 'best of the best' solution to maximize system total energy efficiency and minimize annual total cost.

This study is organized as follows. Section 2 introduces the establishing process of the SOFC-CCHP system model based on multi-software

optimization platform integrating with the SOFC semi-empirical model; Section 3 introduces the device size sensitivity analysis of battery, electrolyzer and SOFC on the CCHP system total energy efficiency and annual total cost, with the contribution rank identification and posteriori multi-criteria decision making for the ‘best of the best’ device sizing. The study results can provide an advanced modeling method of CCHP system simulation and optimization for energy and economic efficiency improvement and provide guidance for the CCHP system design with high techno-economic feasibility for low-carbon district energy transition.

2. Methods

The multi-software optimization platform will be introduced in this section, and the modeling process, operating scenario, control strategy and estimation criteria of the CCHP system will be illustrated. In the end, the logic flow of the multi-objective optimization based on the multi-software optimization platform will be depicted.

The multi-software optimization platform is established with assumptions to ensure the simplification and accuracy:

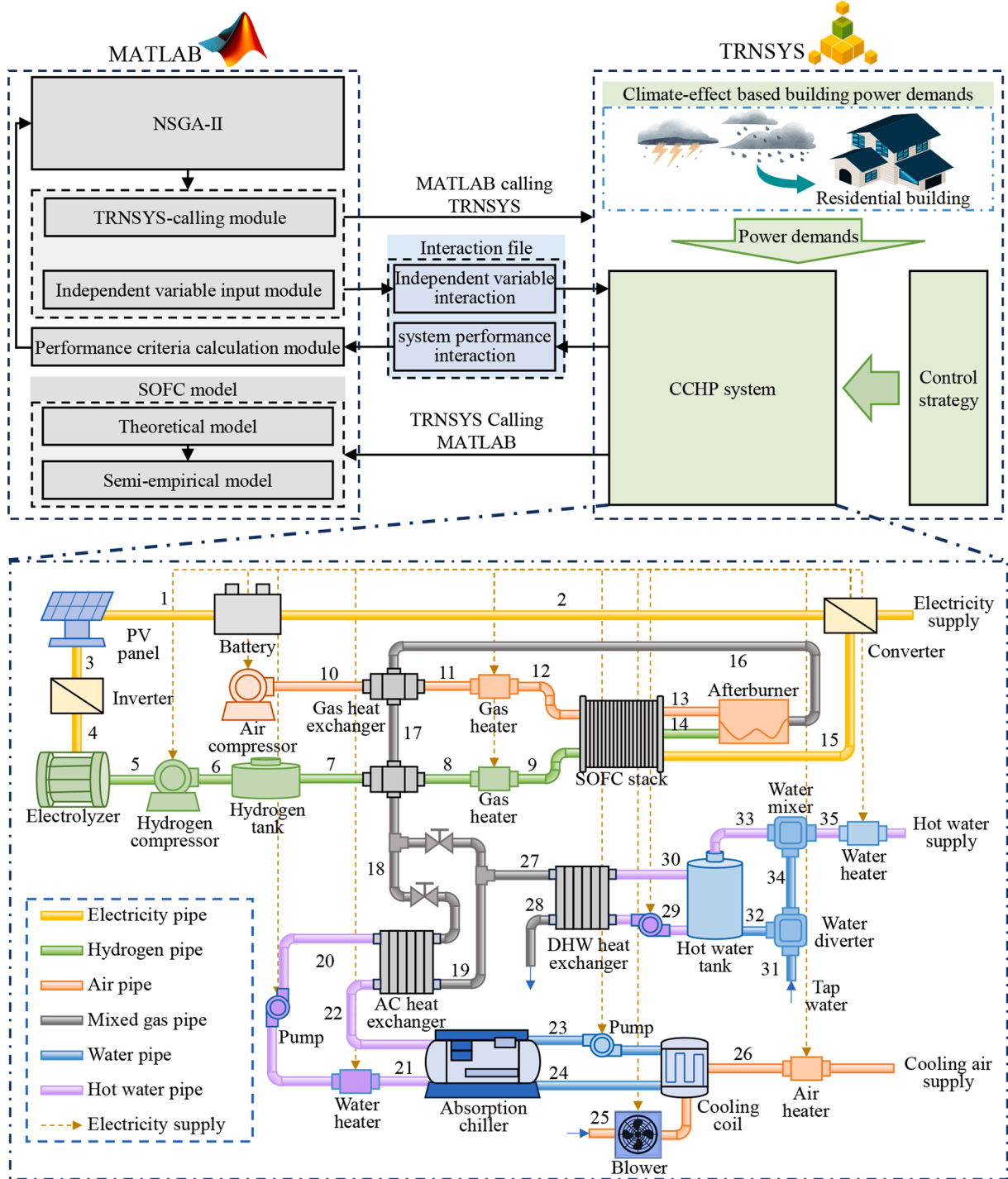


Fig. 1. Multi-software optimization platform based on co-simulation integrating with SOFC semi-empirical model. (Note: NSGA-II, SOFC, CCHP, PV, DHW, AC refer to the non-dominated sorting genetic algorithm II, solid oxide fuel cell, combined cooling, heating and power, photovoltaic, domestic hot water and absorption chiller; the number upon the routes indicate the sequence of the energy/substance flows.).

- (1) Gas species are all ideal gas transporting in the CCHP system;
- (2) Water crossing-membrane effects are not considered in electrolyzer and SOFC;
- (3) Fuel gas in afterburner is burned totally in the excess oxidant mixing;
- (4) The SOFC anode gas consists of H_2 and H_2O , and cathode gas only contains O_2 , N_2 , and H_2O .

2.1. Multi-software optimization platform

Fig. 1 describes the framework design of the multi-software optimization platform, consisting of the cross callings between MATLAB and TRNSYS. As for MATLAB-calling-TRNSYS, the non-dominated sorting genetic algorithm II (NSGA-II) module in MATLAB will control the TRNSYS-calling module to open TRNSYS and then activate the CCHP system model running, and simultaneously input the independent variables into the CCHP system model by interaction files. The CCHP system model in TRNSYS is integrated with the residential building power demands, which is based on the specific climate effects from Meteoronorm database. The control strategy is conducted on CCHP system during the operation process for cost-effectiveness. The CCHP system performances (such as energy efficiency, annual cost, etc) will be output back to NSGA-II algorithm module in MATLAB by interaction files after finishing the scenario operations. As for TRNSYS-calling-MATLAB, the CCHP system model in TRNSYS has integrated with the SOFC model in MATLAB by MATLAB-calling type, and drive the SOFC model calculating in each time step during scenario operation. The operating conditions (inlet gas pressure, temperature, species, load current, etc) will be input to the SOFC theoretical model to calculate the SOFC performance, including voltage, electrical power, thermal power, and etc. Afterwards, the fitting coefficients of the SOFC semi-empirical model will be estimated based on the load current and electrical/thermal power performances. The estimated semi-empirical model of SOFC will then calculate the SOFC stack performance corresponding to the variable load current, and interact with other balance of plant (BOP) models for high-efficiency response. When the operating boundary conditions (inlet gas pressure, stoichiometric ratio, inlet gas fraction and inlet gas temperature) are changed, the fitting coefficients of the semi-empirical model will be updated by once fitting correspondingly.

The CCHP system operation is self-sufficient by only renewable energy support without power grid input, which can be applied in some standalone scenarios, such as the building energy coverage in standalone islands. The lithium iron phosphate (LFP) battery is adopted for energy storage in CCHP system, due to its merits of low cost, high safety and excellent thermal stability [30]. Besides, the proton exchange membrane (PEM) electrolyzer is used to produce H_2 in CCHP system because of its quickly dynamic response and high energy efficiency advantages [31]. As shown in Fig. 1, the PV panels will convert solar energy into electricity, and charge the battery (shown as route 1) for electrical energy storage with charging rate of 0.2C (i.e., 20 % of the capacity for each hour), when the battery state of charge (SOC) is below 60 %. The battery charging will not be stopped until the SOC reaches 95 %. After the battery is completely charged, the surplus electrical energy will be delivered to electrolyzer for H_2 production through the electrolysis process (shown as route 3, 4). Specifically, the input power to the electrolyzer will drive the H_2O in anode to produce O_2 , electrons and protons, and protons will cross the electrolyte to cathode and combine with electrons to produce H_2 . The H_2 from electrolyzer will be stored in hydrogen tank by hydrogen compressor (shown as route 5, 6). Afterwards, the H_2 and air will be supplied by hydrogen tank and air compressor respectively (shown as route 7, 10) to SOFC stack for electricity and heat production. The H_2 and air flow will be heated up in gas heat exchanger (high-temperature exhaust mixed gas) and gas heater (electrical power) successively before entering the SOFC stack (shown as route 8, 9, 11, 12). After crossing the SOFC stack, the surplus H_2 and air

are mixed in afterburner for adequate burning to further improve the temperature (shown as route 13, 14). The high-temperature exhaust mixed gas will cross the cathode and anode gas heat exchanger to heat the inlet H_2 and air flow (shown as route 16, 17). Then it will cross the absorption chiller (AC) heat exchanger (shown as route 18, 19) to heat the generator of absorption chiller by cycle water (shown as route 20 – 22) for cooling energy supply in summer (shown as route 23 – 26). After that the high-temperature mixed gas will cross the domestic hot water (DHW) heat exchanger to heat water (shown as route 27, 28), which is used to supply heating energy to buildings for DHW use and space heating in winter (shown as route 29 – 35). The battery and SOFC will both supply electrical energy to cover the building energy demands and BOP energy consumption (shown as route 2, 15), and their cooperations will be introduced in Section 2.3.

2.2. Combined cooling, heating and power system modeling

In this section, the key process of CCHP system modeling is illustrated, and the detailed process can refer to the previous work [32], including the modeling of battery, electrolyzer, SOFC and other BOP components.

2.2.1. Battery modeling

Battery effective capacity can be calculated by Eq. (1).

$$Ca_{eff} = N_{ba} \bullet Ca_{ini,cell} \bullet RC + P_{ba} \bullet t_{ba} \quad (1)$$

where Ca_{eff} is the effective battery capacity, Wh; N_{ba} is the battery cell number; $Ca_{ini,cell}$ is the battery initial cell capacity, Wh; P_{ba} is the net charging-discharging power, W; t_{ba} is the battery operating duration time, h; RC is the real-time relative capacity, which is related to the charging-discharging cycle number and relative capacity degradation of each cycle, calculated by Eq. (2) [33].

$$RC = 100\% - \sum_{i=1}^{CN} |\Delta RC| \quad (2)$$

where CN is the cycle number; ΔRC is the relative capacity degradation in each cycle, corresponding to the cycle number and each cycle DOD, calculated by Eqs. (3) – (4).

$$\Delta RC = \beta_1 \cdot CN^2 + \beta_2 \cdot CN + \beta_3 \quad (3)$$

$$\beta_j = \sum_{i=0}^8 \gamma_{j,i} \cdot D^i \quad (4)$$

where β_j ($j = 1, 2, 3$) is the coefficients related to the depth of discharge D ; $\gamma_{j,i}$ is the coefficients for polynomial fitting between D and ΔRC coefficients, as shown in Table 1. The battery depth of discharge D can be calculated by Eq. (5).

$$D = SOC_{peak} - SOC_{valley} \quad (5)$$

where SOC_{peak} and SOC_{valley} are the peak value and valley value of the state of charge in a certain cycle number.

2.2.2. Electrolyzer modeling

The electrolyzer produced H_2 mass rate $m_{H_2,ge}$ (kg/s) can be calculated by Eq. (6).

$$m_{H_2,ge} = \frac{I_{el} \cdot N_{el} \cdot M_{H_2}}{n \cdot F} \quad (6)$$

where N_{el} is the electrolyzer cell number; M_{H_2} is the H_2 molar mass, kg/mol; n is the electron number; F is the Faraday constant, 96,485C/mol; I_{el} is the electrolyzer input current (A), calculated by Eq. (7).

$$I_{el} = P_{el} / \left(N_{el} \cdot E_{el,cell}^{eff} \right) \quad (7)$$

where P_{el} is the total input power to electrolyzer, W; $E_{el,cell}^{eff}$ is the

Table 1The coefficients γ_{ji} for polynomial fitting [32].

γ_{ji}	$i = 0$	$i = 1$	$i = 2$	$i = 3$	$i = 4$	$i = 5$	$i = 6$	$i = 7$	$i = 8$
$j = 1$	-7.076×10^{-11}	8.014×10^{-10}	-6.182×10^{-9}	1.406×10^{-8}	0	-3.915×10^{-8}	0	6.358×10^{-8}	-4.086×10^{-8}
$j = 2$	7.654×10^{-7}	-2.129×10^{-6}	2.446×10^{-5}	-5.134×10^{-5}	0	1.742×10^{-4}	0	-3.285×10^{-4}	2.127×10^{-4}
$j = 3$	-0.005	0.002	-0.041	0.0743	0	-0.238	0	0.502	-0.326

electrolyzer effective cell voltage (V), calculated by Eq. (8).

$$E_{el,cell}^{eff} = E_{el,cell} + \Delta E_{el} \quad (8)$$

where $E_{el,cell}$ is the electrolyzer cell voltage, V; ΔE_{el} is the degradation voltage of electrolyzer. Wherein the electrolyzer cell voltage $E_{el,cell}$ can be calculated by Eq. (9).

$$E_{el,cell} = E_{el,re} + E_{el,act} + E_{el,ohm} \quad (9)$$

where $E_{el,re}$ is the reversible potential (V), described as Eq. (10).

$$E_{el,re} = 1.229 - 8.5 \times 10^{-4} \cdot (T_{el} - 298.15) \quad (10)$$

where T_{el} is the operating temperature, 353.15 K [34]. $E_{el,act}$ is the active overpotential (V), calculated by Eq. (11) [35].

$$E_{el,act} = \frac{RT_{el}}{F} \ln \left(\frac{1}{2i_{el,a}} + \sqrt{1 + \left(\frac{I_{el}}{2i_{el,a}} \right)^2} \right) + \frac{RT_{el}}{F} \ln \left(\frac{1}{2i_{el,c}} + \sqrt{1 + \left(\frac{I_{el}}{2i_{el,c}} \right)^2} \right) \quad (11)$$

where R is the gas constant, 8.314 J/(mol·K); $i_{el,a}$ and $i_{el,c}$ are the anode and cathode exchange current density of electrolyzer, A/m².

$E_{el,ohm}$ is the ohmic overpotential (V), calculated by Eq. (12).

$$E_{el,ohm} = I_{el} \cdot \int_0^{d_{el}} \frac{dx}{\sigma(\lambda)} \quad (12)$$

where d_{el} is the thickness of electrolyzer membrane, 5×10^{-5} m [35]; $\sigma(\lambda)$ is the ionic conductivity (S/m), which is calculated by Eq. (13).

$$\sigma(\lambda) = (0.513\lambda - 0.326) \cdot \exp \left[1268 \cdot \left(\frac{1}{303} - \frac{1}{T_{el}} \right) \right] \quad (13)$$

where λ is the membrane water content.

The degradation voltage ΔE_{el} is calculated by Eq. (14) [36].

$$\Delta E_{el} = N_{el} \cdot (t_{mt} \cdot \mu_{mt} + t_{fl} \cdot \mu_{fl} + t_{ct} \cdot \mu_{ct} + t_{fh} \cdot \mu_{fh} + t_{cr} \cdot \mu_{cr}) \quad (14)$$

where t_{mt} , t_{fl} , t_{ct} , t_{fh} , t_{cr} are the operating time duration of the maintenance operation, low power fluctuation operation, constant turning power operation, high power fluctuation operation and constant rated power operation; μ_{mt} , μ_{fl} , μ_{ct} , μ_{fh} , μ_{cr} are the voltage degradation rate in corresponding conditions, as shown in Table 2.

The electrolyzer energy loss includes idling energy loss, auxiliary

Table 2

Voltage degradation rate of electrolyzer during different operation periods [36].

Operation period	Power density range (W/cm ²)	Degradation rate (μ V/h)
Maintenance operation	≤ 0.6	1.5
Low power fluctuation operation	0.6–4	50
Constant turning power operation	4	20
High power fluctuation operation	4–6	66
Constant rated power operation	6	196

energy loss, overpotential energy loss and degradation energy loss, expressed as Eq. (15).

$$W_{el,loss} = W_{el,idling} + W_{el,auxi} + W_{el,E} + W_{el,de} \quad (15)$$

The idling energy loss of electrolyzer $W_{el,idling}$ can be calculated by Eq. (16).

$$W_{el,idling} = \int_0^{8760} V_{el,idling} \cdot I_{el,idling} dt \quad (16)$$

where $V_{el,idling}$ is the electrolyzer idling voltage (V), namely the minimum allowable voltage for activation; $I_{el,idling}$ is the corresponding current under the idling voltage, A. The auxiliary energy loss $W_{el,auxi}$ is the water pump energy consumption for the electrolyzer cooling, which can be output from TNRSYS types. The overpotential energy loss $W_{el,E}$ is calculated by Eq. (17).

$$W_{el,E} = \int_0^{8760} I_{el} \cdot (E_{el,act} + E_{el,ohm}) dt \quad (17)$$

The degradation energy loss $W_{el,de}$ is calculated by Eq. (18).

$$W_{el,de} = \int_0^{8760} I_{el} \cdot \Delta E_{el} dt \quad (18)$$

2.2.3. Solid oxide fuel cell theoretical modeling

The cell number of SOFC affects the electrical/thermal energy efficiency, as shown in Fig. 2(a). Specifically, when the output electrical power demand of SOFC stack is similar, each cell of SOFC stack will hold a large power load if the stack cell number is small. On the contrary, each cell power load will be low if the stack cell number is large. Thus, the SOFC cell power of large cell number design will be lower than that of small cell number design when covering a similar electrical power demand. A lower cell power of SOFC indicates a higher cell voltage and lower overpotential, i.e., the higher electrical energy efficiency and lower thermal energy efficiency. Besides, the performance degradation will promote thermal energy efficiency, as shown in Fig. 2(b). The SOFC contains a lower cell voltage after degradation when covering a similar electrical power demand, indicating that SOFC has a higher overpotential with more heat production. Thus, the SOFC cell number and performance degradation effects are non-negligible on combined heating and power co-supply during the long-time operation.

The SOFC stack electrical power $P_{s,e}$ is calculated by Eq. (19).

$$P_{s,e} = E_{s,cell}^{eff} \cdot i_s \cdot A_s \cdot N_s \quad (19)$$

where $E_{s,cell}^{eff}$ is the SOFC effective cell voltage, V; i_s is the SOFC current density, A/m²; A_s is the active area of SOFC, m²; N_s is the SOFC cell number.

The SOFC thermal power $P_{s,t}$ can be calculated by Eq. (20).

$$P_{s,t} = (1.254 \cdot E_{s,cell}^{eff}) \cdot i_s \cdot A_s \cdot N_s \quad (20)$$

The effective SOFC cell voltage $E_{s,cell}^{eff}$ is described as Eq. (21).

$$E_{s,cell}^{eff} = E_{s,cell} \cdot (1 - \omega_s \cdot t_s) \quad (21)$$

where $E_{s,cell}$ is the SOFC cell voltage, V; ω_s is the SOFC performance degradation rate, %/h; t_s is the SOFC operating time duration, h. Wherein the SOFC cell voltage $E_{s,cell}$ is calculated by Eq. (22).

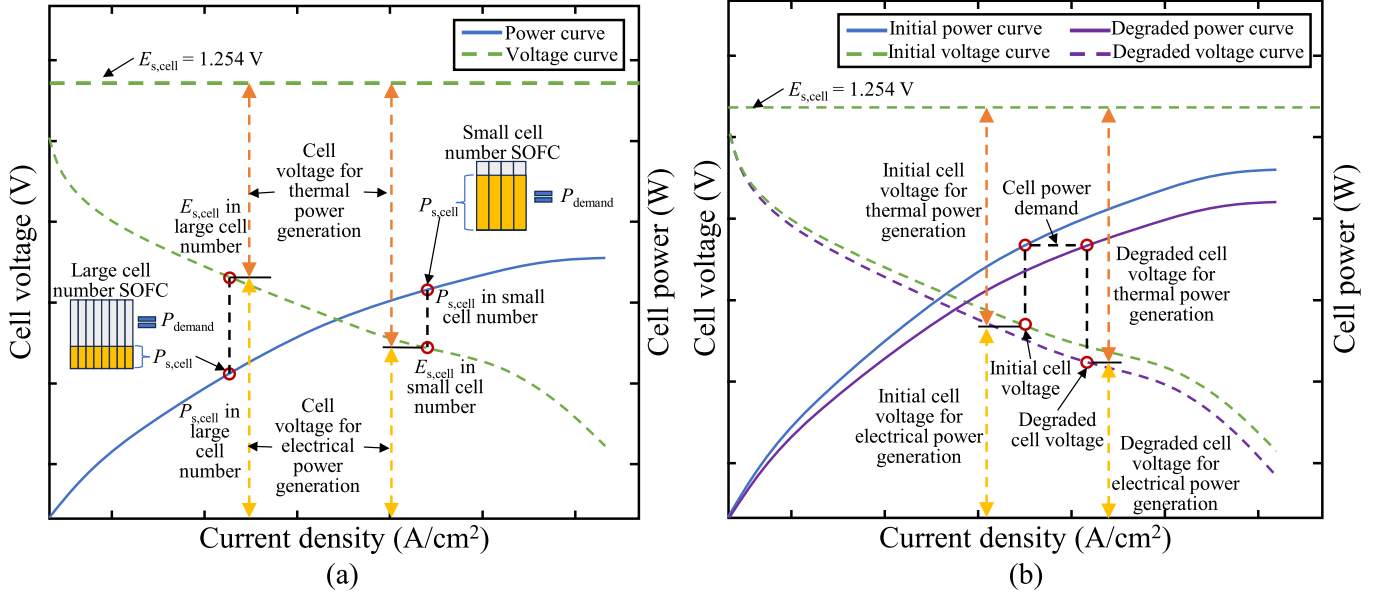


Fig. 2. Effects of SOFC cell number and performance degradation on cell voltage for electrical/thermal power generation: (a) cell number effects; (b) performance degradation effects. (Note: P_{demand} , $P_{\text{s,cell}}$ and $E_{\text{s,cell}}$ refer to the output electrical power demand, SOFC cell power and SOFC cell voltage).

$$E_{\text{s,cell}} = E_{\text{s,ner}} - E_{\text{s,act}} - E_{\text{s,ohm}} - E_{\text{s,con}} \quad (22)$$

where $E_{\text{s,ner}}$ is the Nernst potential, V; $E_{\text{s,act}}$, $E_{\text{s,ohm}}$ and $E_{\text{s,con}}$ refer to the active overpotential, ohmic overpotential, and concentration overpotential, V. The Nernst potential $E_{\text{s,ner}}$ is calculated by Eq. (23).

$$E_{\text{s,ner}} = 1.253 - 2.4516 \times 10^{-4} \cdot T_s \cdot \ln \left(\frac{p_{\text{H}_2} \cdot p_{\text{O}_2}^{0.5}}{p_{\text{H}_2\text{O}}} \right) \quad (23)$$

where p_{H_2} , p_{O_2} , $p_{\text{H}_2\text{O}}$ refer to the gas pressure of H_2 , O_2 and H_2O ; T_s is the SOFC operating temperature, K. The active overpotential $E_{\text{s,act}}$ is calculated by Eq. (24).

$$E_{\text{s,act}} = \frac{RT_s}{F} \cdot \ln \left[\left(\frac{i_s}{2i_{s,a}} \right) + \sqrt{\left(\frac{i_s}{2i_{s,a}} \right)^2 + 1} \right] + \frac{RT_s}{F} \ln \left[\left(\frac{i_s}{2i_{s,c}} \right) + \sqrt{\left(\frac{i_s}{2i_{s,c}} \right)^2 + 1} \right] \quad (24)$$

where $i_{s,a}$ and $i_{s,c}$ are the SOFC anode and cathode exchange current density, A/m^2 . The ohmic overpotential $E_{\text{s,ohm}}$ is calculated by Eq. (25).

$$E_{\text{s,ohm}} = i_s \cdot \left(\frac{d_a}{\kappa_a} + \frac{d_c}{\kappa_c} + \frac{d_{\text{ele}}}{\kappa_{\text{ele}}} + \frac{d_{\text{int}}}{\kappa_{\text{int}}} \right) \quad (25)$$

where d_a , d_c , d_{ele} , d_{int} are the thickness of anode, cathode electrolyte and interconnector of single cells, m, 4.00×10^{-4} m, 4.00×10^{-5} m, 2.00×10^{-5} m, 8.50×10^{-5} m, respectively [37,38]; κ_a , κ_c , κ_{ele} , κ_{int} are electron conductivity of anode, cathode, electrolyte and interconnector. The concentration overpotential $E_{\text{s,con}}$ is calculated by Eq. (26).

$$E_{\text{s,con}} = \frac{-RT_s}{n_{\text{H}_2} F} \ln \left(\frac{1 - i_s / i_{\text{lim,H}_2}}{1 + i_s / i_{\text{lim,H}_2\text{O}}} \right) + \frac{RT_s}{n_{\text{O}_2} F} \ln \left(\frac{1}{1 - i_s / i_{\text{lim,O}_2}} \right) \quad (26)$$

where $i_{\text{lim,H}_2}$, $i_{\text{lim,H}_2\text{O}}$, and $i_{\text{lim,O}_2}$ are the limiting current density of H_2 , H_2O and O_2 .

The Performance degradation rate ω_s (%/h) is calculated by Eq. (27) [39].

$$\omega_s = \frac{0.59 \cdot \text{mf}_{\text{H}_2} + 0.74}{1000 \left[1 + \exp \left(\frac{T_s - 1087}{22.92} \right) \right]} \left[\exp \left(\frac{2.64 i_s}{10000} \right) - 1 \right] \quad (27)$$

where mf_{H_2} is the mass fraction of H_2 consumption.

The thermal energy conservation of SOFC stack can be described by Eq. (28).

$$p_{t,a}^{\text{in}} + p_{t,c}^{\text{in}} + P_{s,t} = p_{t,a}^{\text{out}} + p_{t,c}^{\text{out}} + P_{t,\text{loss}} \quad (28)$$

where $p_{t,a}^{\text{in}}$, $p_{t,c}^{\text{in}}$, $p_{t,a}^{\text{out}}$, $p_{t,c}^{\text{out}}$, and $P_{t,\text{loss}}$ refer to the anode inlet thermal power, cathode inlet thermal power, anode outlet thermal power, cathode outlet thermal power, and thermal power loss in ambience. Wherein the thermal power loss $P_{t,\text{loss}}$ in ambience can be calculated by Eq. (29).

$$P_{t,\text{loss}} = \left| h \int_0^S (T_{\text{am}} - T_s) dS + \sigma \cdot \gamma \cdot \int_0^S (T_{\text{am}}^4 - T_s^4) dS \right| \quad (29)$$

where h is the convective heat transfer coefficient, $9 \text{ W}/(\text{m}^2 \cdot \text{K})$ [4]; S is the surface area of the SOFC stack; σ is the Stefan-Boltzmann constant, $5.67 \times 10^{-8} \text{ W}/(\text{m}^2 \cdot \text{K}^4)$; γ is the emissivity of the SOFC surface, 0.5 [40]; T_{am} is the ambient temperature, K.

In order to improve the authenticity and detailedness of SOFC model, the outlet gas temperature of SOFC stack is calculated by nonlinear conservation solving method of least square method (LSM) based on the thermal energy conservation, which can be described by Eq. (30).

$$T_{\text{gas}}^{\text{out}} = \text{FSOLVE} \left(p_{t,a}^{\text{in}} + p_{t,c}^{\text{in}} + P_{s,t} - p_{t,a}^{\text{out}} - p_{t,c}^{\text{out}} - P_{t,\text{loss}} = 0 \right) \quad (30)$$

where $T_{\text{gas}}^{\text{out}}$ is the outlet gas temperature of the SOFC stack, K; FSOLVE refer to the LSM solver, and the numerical relationships between gas temperature and thermal powers can refer to the previous work [32].

The anode inlet mass flow rate of SOFC stack m_a (kg/s) is calculated as Eq. (31).

$$m_a = \frac{i_s \cdot A_s \cdot N_s \cdot M_{\text{H}_2} \cdot \alpha_a}{n_{\text{H}_2} \cdot F \cdot \text{mf}_{\text{H}_2}^{\text{in}}} \quad (31)$$

where α_a is the anode stoichiometric ratio; $\text{mf}_{\text{H}_2}^{\text{in}}$ is the inlet H_2 mass fraction.

The cathode inlet mass flow rate of SOFC stack m_c (kg/s) is calculated as Eq. (32).

$$m_c = \frac{i_s \cdot A_s \cdot N_s \cdot M_{O_2} \cdot \alpha_c}{n_{O_2} \cdot F \cdot mf_{O_2}^{in}} \quad (32)$$

where α_c is the cathode stoichiometric ratio; $mf_{O_2}^{in}$ is the inlet O_2 mass fraction.

The battery, electrolyzer and SOFC parameter settings of structures, materials and operating conditions are illustrated in Table 3.

2.2.4. Solid oxide fuel cell semi-empirical modeling

Due to the time-consuming characteristic of nonlinear conservation solving to calculate SOFC stack performance in each time step during year-scale scenarios, a semi-empirical model of SOFC stack is developed to surrogate the theoretical model, for high-efficiency interaction response with BOP models during the system operation, aiming to accelerate the multi-objective optimization process. The semi-empirical cell voltage $E_{s,cell}^{sem}$ can be calculated by Eq. (33).

$$E_{s,cell}^{sem} = 1.254 \cdot E_{s,act}^{sem} - E_{s,ohm}^{sem} - E_{s,con}^{sem} \quad (33)$$

where $E_{s,act}^{sem}$, $E_{s,ohmic}^{sem}$, and $E_{s,con}^{sem}$ refer to the semi-empirical active overpotential, semi-empirical ohmic overpotential, and semi-empirical concentration overpotential, calculated by Eqs. (34) – (36) [19].

$$E_{s,act}^{sem} = \delta_1 \cdot \ln(i_s) \quad (34)$$

$$E_{s,ohm}^{sem} = \delta_2 \cdot i_s \quad (35)$$

$$E_{s,con}^{sem} = \delta_3 \cdot \exp(\delta_4 \cdot i_s) \quad (36)$$

where δ_1 , δ_2 , δ_3 , δ_4 indicate the fitting coefficients changed with the operating condition variations.

The semi-empirical electrical power $P_{e,sem}$ is calculated by Eq. (37).

$$P_{e,sem} = E_{s,cell}^{sem} \cdot (1 - \omega_s \cdot t_s) \cdot i_s \cdot A_s \cdot N_s \quad (37)$$

The semi-empirical thermal power $P_{t,sem}$ is calculated by Eq. (38).

$$P_{t,sem} = [1.254 \cdot E_{s,cell}^{sem} \cdot (1 - \omega_s \cdot t_s)] \cdot i_s \cdot A_s \cdot N_s \quad (38)$$

The outlet gas temperature $T_{sem,gas}^{out}$ is calculated by Eq. (39).

$$T_{sem,gas}^{out} = T_{gas}^{in} + \frac{P_{t,sem}}{\delta_5 \cdot \bar{C}_{gas} \cdot \bar{m}_{gas}} \quad (39)$$

where T_{gas}^{in} is the inlet gas temperature of SOFC stack, K; δ_5 is the fitting coefficient; $\bar{C}_{gas}$ and $\bar{m}_{gas}$ refer to the average specific heat capacity (J/

(kg·K)) and average mass flow rate (kg/s) of the inlet and outlet mixed gas of SOFC stack.

2.2.5. Balance of plant component modeling

The BOP components of CCHP system include PV panels, inverter, compressor, heat exchanger, heater, afterburner, pump, absorption chiller, air heater, hot water tank, water diverter, water mixer, and building power demand type. The afterburner model is established based on the energy conservation solving, and other BOP components are selected from TRNSYS types. The modeling process and parameter settings of BOP component types can refer to the previous work [32].

2.3. Scenario and control strategy

In this section, the operating scenario and corresponding control strategy of the CCHP system will be described, which will operate in a residential building in Beijing, China.

2.3.1. Scenario

The CCHP system will be applied in a residential building located in Beijing, China (temperate area). The DHW supply temperature in buildings is set as 60 °C, and the temperature setting of cooling air for space cooling is 22 °C. Fig. 3 exhibits the electrical power demand, DHW heating power demand, space cooling power demand, and space heating power demand of the residential building during a whole year [41]. Specifically, the DHW heating power demand is stable around the year. The electrical power will be much higher during winter (0–1460 h, 8030–8760 h) and summer (3650–5840 h), with peak value of 10.005 kW. The space cooling power demand and space heating power demand only appear in summer and winter, with peak values of 8.006 kW and 8.331 kW, respectively. In order to guarantee the computational efficiency, the time step size of the simulation is set as 0.25 h. Thus, the transient behaviors of the devices maintaining within 0.25 h will not be considered in the simulation.

2.3.2. Control strategy

The control strategy of the CCHP system has distinguished different dominated power-supply devices (battery and SOFC) during daily operation, and taking two days in winter and summer respectively for instances, Fig. 4 illustrates the daily power demands of the residential building and operation conditions of battery and SOFC. The peak power demand period during the whole day will be covered by SOFC energy supply, i.e., 6:00 am – 10:00 am and 17:00 pm – 2:00 am, and the valley power demand period will be covered by battery energy supply, including periods of 2:00 am – 6:00 am and 10:00 am – 17:00 am. Besides, the start-up period of SOFC stack will be set as 2 h [42] (4:00 am – 6:00 am and 15:00 am – 17:00 am), during which the battery will supply the BOP component operation and assist the SOFC stack to heat up from the ambient temperature (non-working state) to setpoint temperature (preparing-working state). The period-classified power supply by SOFC and battery alternating domination is able to achieve more cost-effective performance, due to the alternately intermittent rest of the power supply devices, which can reduce by 11.83 % annual total cost compared to the operation strategy of both battery (only BOP energy supply) and SOFC (only building demand supply) working continuously without rest [32].

2.4. Multi-criteria performance estimation

The estimation criteria of the CCHP system performance include total energy efficiency and annual total cost.

The system total energy efficiency f_{sys} can be calculated by Eq. (40) [43].

$$f_{sys} = \frac{\int_0^{8760} (P_e^{sys} + P_{heating}^{sys} + P_{cooling}^{sys}) dt}{\int_0^{8760} (P_{ra} A_{PV} N_{PV}) dt + W_{ba,ini} + W_{H_2,ini}} \quad (40)$$

Table 3

The parameter settings of the key devices [32].

Devices	Parameter settings	Values
battery	Cell capacity (Wh)	400
	Active area (m ²)	0.125
Electrolyzer	Operating temperature (K)	353.15
	minimum allowable cell voltage (V)	1.4
	Membrane thickness (m)	5×10^{-5}
	membrane interface water content λ	14
SOFC	Active area (m ²)	0.25
	Anode thickness (m)	4.00×10^{-4}
	Cathode thickness (m)	4.00×10^{-5}
	Electrolyte thickness (m)	2.00×10^{-5}
	Interconnector thickness (m)	8.50×10^{-5}
	Anode inlet gas temperature (K)	873.15
	Cathode inlet gas temperature (K)	873.15
	Anode inlet gas pressure (atm)	2
	Cathode inlet gas pressure (atm)	2
	Anode inlet H ₂ mass fraction (%)	50
	Anode inlet H ₂ O mass fraction (%)	50
	Cathode air mass fraction (%)	100
	Anode stoichiometric ratio	1.1
	Cathode stoichiometric ratio	5.0

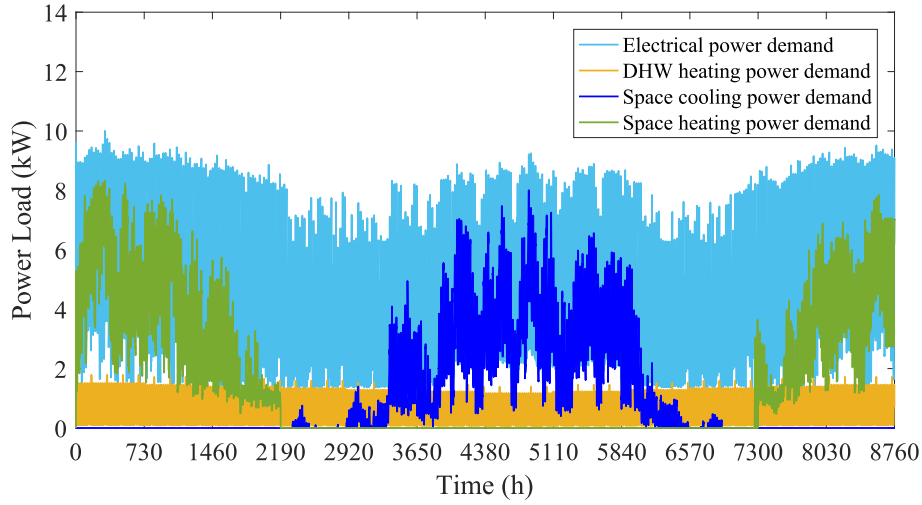


Fig. 3. Power load of residential building in Beijing including electric power, DHW heating power, cooling load and heating load. (Note: DHW refer to the domestic hot water).

where f_{sys} is the system energy efficiency; P_e^{sys} is the system electric power output, W; $P_{heating}^{sys}$ is the system heating power output, W; $P_{cooling}^{sys}$ is the system cooling power output, W; P_{ra} is the solar radiation power density, W/m²; A_{pv} is the area of PV panel, m²; N_{pv} is the installed PV panel number; $W_{ba,ini}$ is the battery initial energy consumption, i.e., the capacity difference between year-end condition and initial condition, dependent on the net charging-discharging power P_{ba} , which can be calculated by Eq. (41).

$$W_{ba,ini} = \int_0^{8760} P_{ba} \cdot t_{ba} dt \quad (41)$$

$W_{H_2,ini}$ is the H₂ initial energy consumption, calculated by Eq. (42).

$$W_{H_2,ini} = m_{H_2}^{net} \cdot LHV_{H_2} \quad (42)$$

where LHV_{H_2} is the low heat value of H₂, J/kg; $m_{H_2}^{net}$ is the H₂ net consumption mass (kg), i.e., the H₂ mass difference in the hydrogen tank between year-end condition and initial condition, calculated by Eq. (43).

$$m_{H_2}^{net} = \int_0^{8760} (m_{H_2}^{co} - m_{H_2}^{ge}) dt \quad (43)$$

where $m_{H_2}^{co}$ and $m_{H_2}^{ge}$ is the real-time H₂ consumption rate and generation rate, kg/s.

The system annual total cost $Cost_{total}$ can be calculated by Eq. (44).

$$Cost_{total} = Cost_{ini} \cdot CR + Cost_{main} + Cost_{dep} \quad (44)$$

where $Cost_{ini}$ is the initial investment cost, as shown in Table 4; CR is the capital recovery factor, calculated by Eq. (45).

$$CR = \frac{r \cdot (r + 1)^L}{(r + 1)^L - 1} \quad (45)$$

where r is interest rate, 5 %; L is the CCHP system lifetime, 20 years [44].

$Cost_{main}$ is the maintenance cost (\$), which is dependent on the initial investment cost, calculated by Eq. (46).

$$Cost_{main} = \zeta \cdot Cost_{ini} \quad (46)$$

where ζ is the maintenance coefficient, 6 % [43].

$Cost_{dep}$ is the annual total depreciation cost (\$), indicating the degradation levels of battery, electrolyzer and SOFC stack, which can be calculated by Eq. (47).

$$Cost_{dep} = \frac{RC_{de}}{RC_{de,max}} \cdot Cost_{ini,ba} + \frac{\Delta E_{el}}{\Delta E_{el,max}} \cdot Cost_{ini,el} + \frac{\Delta E_s}{\Delta E_{s,max}} \cdot Cost_{ini,s} \quad (47)$$

where $RC_{de,max}$, $\Delta E_{el,max}$ and $\Delta E_{s,max}$ refer to the maximum allowable degradation capacity of battery, 20 % [45], the maximum allowable degradation voltage of electrolyzer, 0.28 V [46], and the maximum allowable degradation voltage of SOFC, 0.2 V [14]; $Cost_{ini,ba}$, $Cost_{ini,el}$, and $Cost_{ini,s}$ are the initial investment costs of battery, electrolyzer and SOFC respectively, as shown in Table 4.

2.5. Multi-objective optimization by co-simulation

The key device sizes (i.e., the cell numbers of battery, electrolyzer and SOFC) of the CCHP system will be optimized to achieve the optimal annual total cost and total energy efficiency based on the multi-software optimization platform, described as Eq. (48).

$$[\min(Cost_{total}), \max(f_{sys})] = \text{function}(N_{ba}, N_{el}, N_s) \quad (48)$$

Fig. 5 shows the logic flow chart of the multi-objective optimization process. The original population consists of original battery, electrolyzer and SOFC cell number database, and corresponding CCHP system performances (annual total cost and total energy efficiency). The original population is established by co-simulation. Specifically, the original database of battery, electrolyzer and SOFC cell number will be generated randomly in the main function, and then input to the CCHP system to reconstruct the numerical models of battery, electrolyzer and SOFC. After finishing the calculation of the whole-year scenario operation, the performances of CCHP system will be delivered back to the main function, including the degradation level, total input energy, fuel consumption mass, operating time duration, etc. Afterwards, the criteria of annual total cost and total energy efficiency will be calculated. After the original population establishment, the self-checking program will be conducted, and namely the operating duration time of CCHP system will be checked to figure out whether the operation is effective. The operating time duration lower than 8760 h (one year) indicates that the operation has been stopped halfway, which means that the CCHP system model cannot be solved in such device sizes, and the ineffective device sizes will be output before the program stops. If the operating time duration is equal to 8760 h, the original population will be executed of cross calculation, mutation calculation, non-dominated sorting, crowding distance calculation, and Pareto ranking successively. Each time of CCHP system model solving will activate the self-checking program to guarantee the effectiveness of the population database, including the

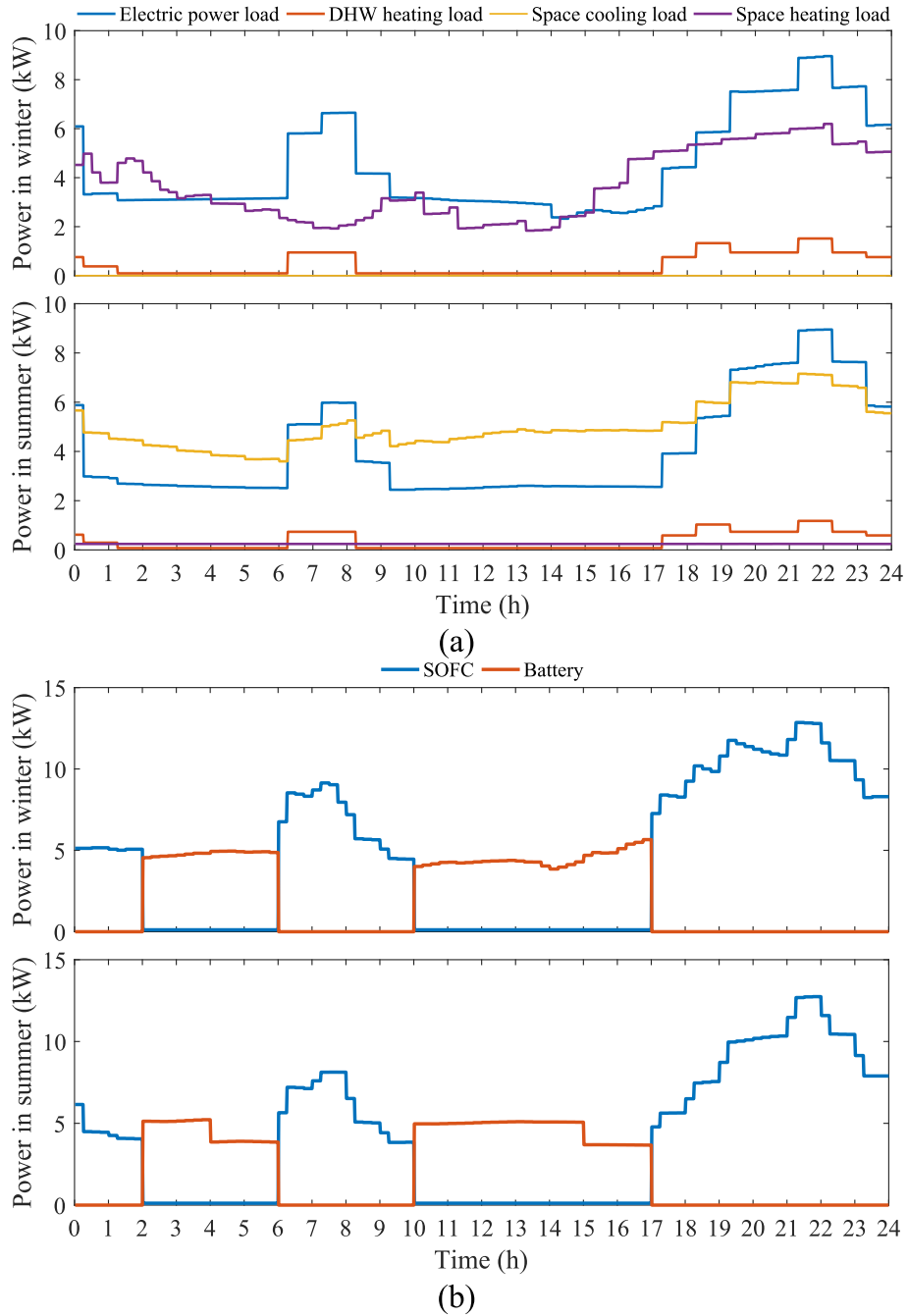


Fig. 4. Daily power demands of the residential building and energy-supply device operation conditions: (a) daily power demands of the residential building; (b) energy-supply device operation conditions.

cross calculation and mutation calculation. The loop will not be stopped until the setting number of iterations is finished.

ANOVA analysis can be used to quantify the influence ability of input factors on the output objectives by significance test [4], which is effective to identify the contribution ranks of different device sizes to the CCHP system performance. The process can be illustrated as:

$$SST = SSF + SSE \quad (49)$$

$$DFT = DFF + DFE \quad (50)$$

$$MSF = SSF/DFF \quad (51)$$

$$MSE = SSE/DFE \quad (52)$$

$$F = MSF/MSE \quad (53)$$

where SST is the sum of squares of total objective values; SSF is the sum of squares for factor variation influences among different groups; SSE is the sum of squares of error effects within group; DFT is the total freedom degree; DFF is the freedom degree of different factor groups; DFE is the freedom degree of error effects in group; MSF is the mean square sum of different factor groups; MSE is the mean square sum of error effects in group; F is the F value. The much higher of F value, the much more significant of input factors on the output objectives. In present research, the ANOVA analysis is used to rank the significance of the key device sizing on system techno-economic performances, and give a guidance for the priority of the key device sizing design.

The 'best of the best' optimal cases can be selected by normalized

Table 4

Initial investment costs of the CCHP system.

Components	Initial investment cost (\$)
Battery [47]	$209 \times C_{\text{ba}}$
Electrolyzer [44]	$1000 \times P_{\text{el}}$
SOFC [48]	$8194 \times P_s$
Afterburner [49]	$\frac{46.08 \times m_{\text{in}}}{0.995-0.98} \times (1 + e^{0.018T_{\text{out}}-26.4})$
PV panel [50]	$82.615 \times A_{\text{PV}}$
Hydrogen tank [51]	$734 \times V_{\text{tank}}$
Pump [52]	$200 \times P_{\text{pump}}^{0.65}$
Compressor [13]	$91562 \times \left(\frac{P_{\text{com}}}{4555}\right)^{0.67}$
Inverter [52]	$10^5 \times \left(\frac{P_{\text{inverter}}}{500}\right)^{0.7}$
Heat exchanger [13]	$130 \times \left(\frac{A_{\text{HE}}}{0.093}\right)^{0.78}$
Heater [44]	$0.3 \times m$
Hot water tank [53]	$0.954V_{\text{tank}} + 832.7$
Absorption chiller [44]	$Q_{\text{cooling}}^{0.67}$

Euclidean distance judging method based on the pareto front results. The normalized Euclidean distance can be calculated by Eq. (54).

$$E_{\text{dis}} = \sqrt{\sum_{i=1}^n (x_{N,i} - x_{N,i}^*)^2} \quad (54)$$

where E_{dis} refers to the normalized Euclidean distance; n is the dimension number of the study cases; $x_{N,i}$ is the normalized study case coordinates; $x_{N,i}^*$ is the normalized optimal case coordinates in theory. The normalized study case coordinates can be calculated by Eq. (55).

$$x_{N,i} = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}} \quad (55)$$

where x_i , $x_{\max}$ and $x_{\min}$ refer to the study case coordinates, study case maximum coordinate value and study case minimum coordinate value.

3. Results and discussions

In this section, the SOFC semi-empirical model is validated. The size effects of the key device (battery, electrolyzer and SOFC) are analyzed on their degradation level, system annual total cost and system total energy efficiency. The optimized key device sizes and their contribution ratios are identified based on optimal annual total cost and total energy efficiency. The battery analyzed size range is set within cell number of 180–960 to independently operate for 4–22 h before charging. In order to control the working power density below 6 W/cm^2 [36] and cell voltage over the idling value of 1.4 V, the electrolyzer size range is set from 8 to 80 cell number. The SOFC analyzed size is set within 15–30 cell number to keep the cell voltage within 0.6–0.85 V in the scenario to prevent from destructive degradation.

3.1. Model validation

The theoretical model and semi-empirical model of SOFC will be validated by experiments in this section. Considering the operating temperature of SDC-based SOFC (500 °C–800 °C [6]), and large energy consumption in long-time scenario operation of CCHP system for high-temperature stabilization, the operating temperature of the SOFC will be set as 600 °C in present research. The SOFC semi-empirical model are validated by comparing with the experimental data [38], in which the boundary conditions are set as 0.05 kg/s mass flow rate containing vol. 90 % H_2 & vol. 10 % H_2O in anode and vol. 100 % air in cathode under the operating temperatures of 550 °C and 600 °C. In order to keep pace with the experimental conditions, and verify the generalization of the SOFC models, the boundary condition settings of the SOFC models are the same as the experiments. The semi-empirical model shows a high accuracy, as shown in Fig. 6. For reference, the R^2 of semi-empirical

model data and theoretical model data fitting are 1.000 in 550 °C condition and 0.999 in 600 °C condition, indicating the surrogate feasibility of semi-empirical model to the theoretical model. Besides, the R^2 between semi-empirical model and experiment results reach 0.959 in 550 °C condition and 0.909 in 600 °C condition, which shows a reliable predicting ability of the SOFC model in a large range of operating temperature conditions.

3.2. System energy efficiency analysis

This section will introduce the effect analyses of battery, electrolyzer and SOFC size designs on the total energy efficiency of the CCHP system.

3.2.1. Battery effects

The annual CCHP system total energy efficiency is dependent on the battery initial energy consumption $W_{\text{ba,ini}}$ (calculated by Eq. (41)) and H_2 net consumption mass $m_{\text{H}_2}^{\text{net}}$ (calculated by Eq. (43)). When the H_2 production mass by electrolyzer is larger than the consumption mass by SOFC, the surplus net produced H_2 will be stored in the hydrogen tank (i. e., $m_{\text{H}_2}^{\text{net}}$ is minus in Eqs. (40)–(43)). The more net H_2 production mass for storage indicates the more solar energy converting into H_2 energy with less energy loss during the conversion process, leading to a much higher system total energy efficiency.

The total energy efficiency of the CCHP system is fluctuated in a limited range without any distinct trend with battery size variation, as shown in Fig. 7(a). Specifically, the total energy efficiency is variable within a small range of 64.888 %–64.936 %, when the battery cell number increases from 180 to 960. The fluctuation of total energy efficiency mainly results from the variable battery initial energy consumption, compared to the stable variation of H_2 net production mass stored in hydrogen tank. In detail, the battery initial energy consumption fluctuates in the range of $6.235 \times 10^7 \text{ J}$ to $3.642 \times 10^8 \text{ J}$ without any variation trends along with the cell number increase. It results from the different charging-discharging cycle times and energy of the battery, as shown in Fig. 7(b), which exhibits the battery charging powers in April with cell number of 540 and 600 for examples. Taking one week for instance, the battery will be charged on Tuesday, Wednesday, Friday and Sunday in cell number of 540, with whole-year total charging energy of $6.049 \times 10^{10} \text{ J}$. However, the battery will be charged on Monday, Wednesday, Thursday and Saturday in cell number of 600, with whole-year total charging energy of $6.077 \times 10^{10} \text{ J}$. It is because that the battery charging from intermittent solar energy is unavailable during nighttime operation, and it leads to a different DOD and discharging energy before sunrise. Therefore, the battery sizing designs play a limited effect on the total energy efficiency of the CCHP system.

3.2.2. Electrolyzer effects

The electrolyzer sizing design will affect the H_2 production directly, and meanwhile, the total energy efficiency and H_2 net production mass of the CCHP system will be improved when the electrolyzer size is enlarged, as shown in Fig. 8(a). To be more specific, the system total energy efficiency will be improved from 64.966 % to 78.601 % (by 13.635 %), and H_2 net production mass will be raised from 18.960 kg to 579.117 kg (by 560.157 kg), along with the electrolyzer cell number increase from 8 to 80. The underlying mechanism is the reduction of electrolyzer energy losses in its size enlargement, as shown in Fig. 8(b). For example, the total energy loss of electrolyzer (sum of idling energy loss, auxiliary energy loss, overpotential energy loss and degradation energy loss) will decrease from $1.742 \times 10^{11} \text{ J}$ to $1.043 \times 10^{11} \text{ J}$ with the cell number rising from 8 to 80. The reason is that the input power held by each electrolyzer cell will decline under the size enlargement, leading to a much lower cell voltage when holding a similar input power, which indicates a lower overpotential loss. Therefore, the enlargement of electrolyzer cell number from 8 to 80 will improve the total energy efficiency of the CCHP system by 13.635 % and H_2 net production mass by

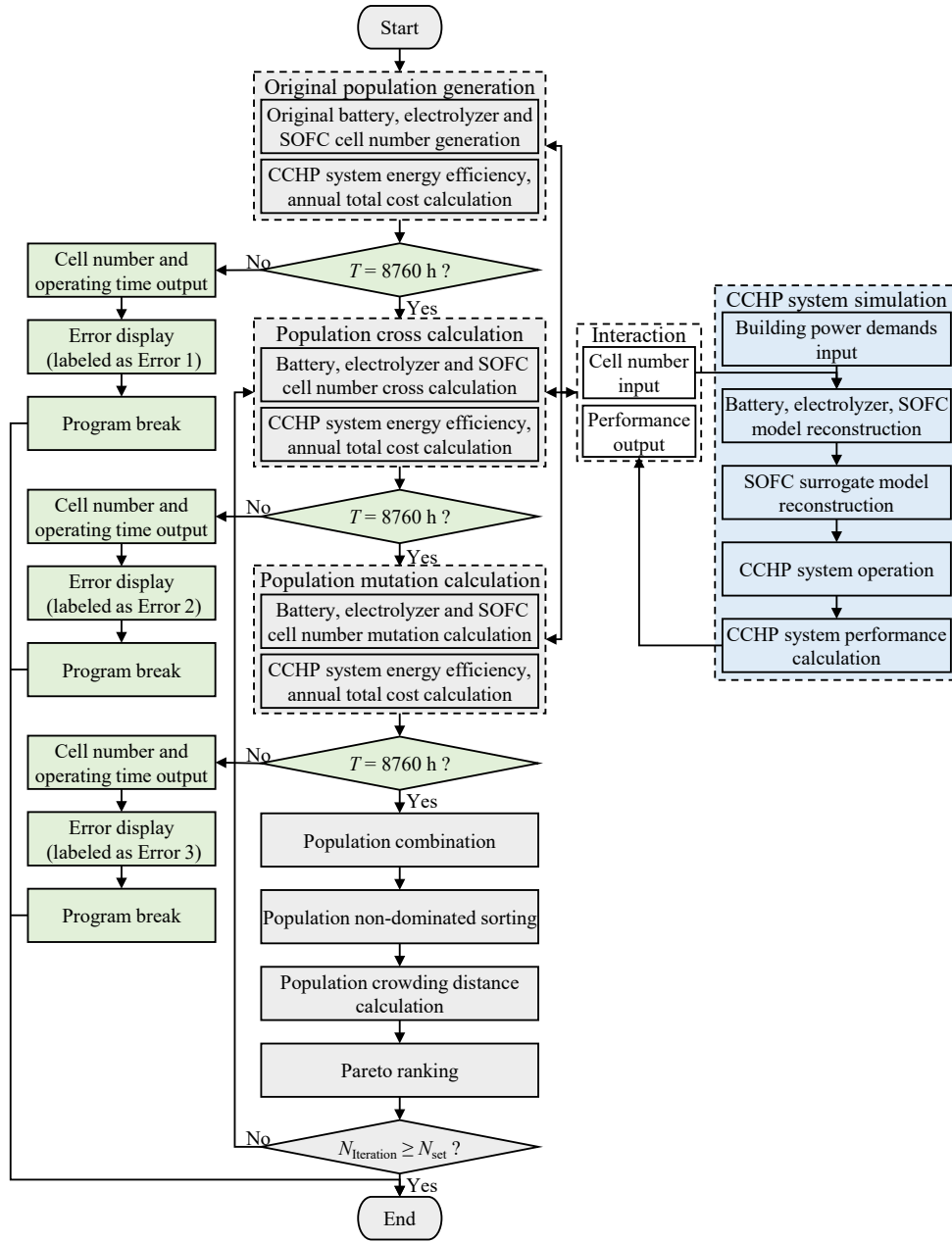


Fig. 5. Logic flow chart of multi-objective optimization based on the multi-software optimization platform.

560.157 kg.

3.2.3. SOFC effects

The electrical energy and thermal energy converted from H_2 energy will be supplied by SOFC and afterburner. As shown in Fig. 9, with the increase in SOFC cell number, the output electrical energy efficiency of H_2 energy will be improved, but the output thermal energy efficiency of H_2 energy will be reduced. To be more specific, the output electrical energy efficiency will be improved from 65.663 % to 70.659 % (by 4.996 %), but the output thermal energy efficiency will decrease from 26.033 % to 23.678 % (by 2.355 %), when the SOFC cell number increases from 15 to 30. The underlying mechanism is related to the different SOFC cell voltage. Specifically, the increase of SOFC cell number will reduce the power output by each cell, leading to a higher cell voltage of SOFC, which indicates a higher electrical energy conversion efficiency and lower thermal energy conversion efficiency, as shown in Fig. 2.

Besides, the output electrical energy is increased but thermal energy will be reduced with the increase in SOFC cell number. For reference, the SOFC electrical energy will increase from 1.137×10^{11} J to 1.165×10^{11} J, while the thermal energy will decrease from 4.508×10^{10} J to 3.903×10^{10} J. The reason is similar to the energy conversion efficiency with the same variation trends.

The enlargement of SOFC cell number will play a positive effect on the CCHP system total energy efficiency and H_2 net production mass, as shown in Fig. 10. Specifically, the system total energy efficiency will be improved from 65.423 % to 67.614 % (by 2.191 %), and the H_2 net production mass will be improved from 47.535 kg to 111.115 kg (by 63.580 kg), with the increase of SOFC cell number from 15 to 30. This results from the higher utilization efficiency of H_2 energy with the increase in SOFC cell number. Specifically, the electrical energy efficiency is enhanced with the SOFC cell number rise, indicating that SOFC will consume less H_2 mass when covering the same electrical energy demand. Meanwhile, the produced thermal energy is enough to cover both

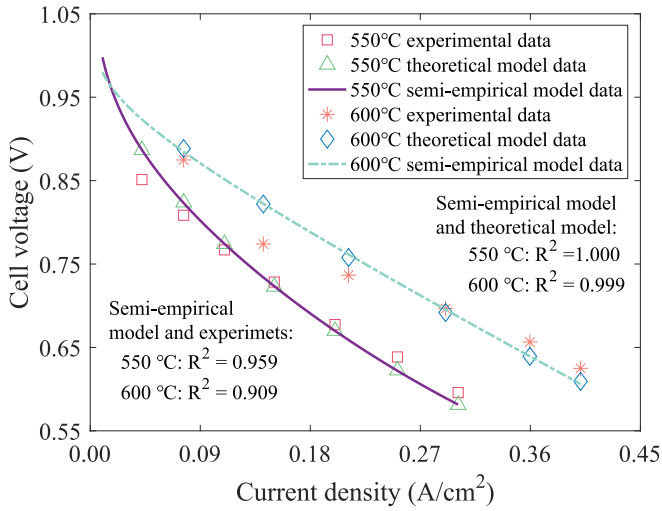


Fig. 6. Comparisons of SOFC experimental data, theoretical model data and semi-empirical model data under operating temperature of 550 °C and 600 °C. (Note: SOFC refers to the solid oxide fuel cell; SOFC data comparisons are under boundary conditions of vol. 90 % H₂ & vol. 10 % H₂O in anode and vol. 100 % air in cathode, and the experiment data refers to [38]).

heating and cooling demands without requiring more H₂ mass consumptions, even if the thermal conversion efficiency decreases. Therefore, the SOFC size enlargement from cell number of 15 to 30 will improve the electrical energy efficiency by 4.996 %, system total energy efficiency by 2.194 % and H₂ net production mass by 63.580 kg.

In addition, comparing Fig. 8(a) to Fig. 10, the trends of hydrogen net production mass and system total energy efficiency with the electrolyzer and SOFC cell number rise are closely similar. Specifically, the hydrogen net production mass is increased by 560.157 kg and 63.580 kg under the cell number rise of electrolyzer (from 8 to 80) and SOFC (from 15 to 30), respectively. The system total energy efficiency is also increased by 13.635 % and 2.191 % by the same size enlargement of electrolyzer and SOFC, respectively. The underlying mechanism is the overpotential mitigation of both electrolyzer and SOFC with their size enlargement. To be more specific, the power and current held by each cell of electrolyzer and SOFC will be reduced with their cell number rise when covering the similar output power load. According to the Tafel equation and Ohm's law, the lower current held by each cell, the less active overpotential loss and ohmic overpotential loss of electrolyzer and SOFC, leading to a higher conversion efficiency of H₂ production and a higher energy efficiency of the CCHP system.

Overall, the size enlargement of battery has little effect on the CCHP system total energy efficiency, while the electrolyzer size enlargement of cell number from 8 to 80 will improve the total energy efficiency by

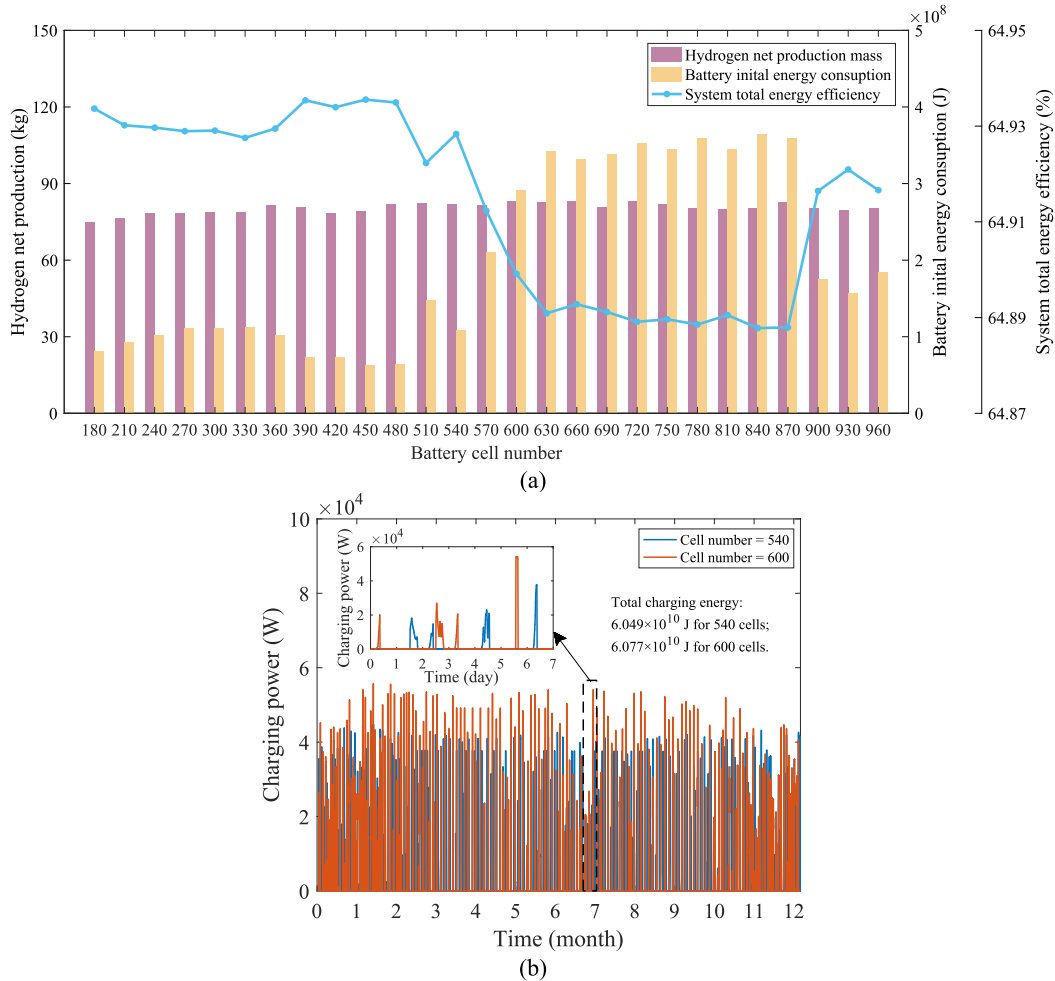


Fig. 7. (a) H₂ net production mass ($|m_{H_2}^{net}|$), battery initial energy consumption ($W_{ba,ini}$) and total energy efficiency (f_{sys}) of the CCHP system during a whole year in variable battery cell numbers; (b) battery charging power with cell number designs of 540 and 600 during a whole year. (Notes: H₂ net production mass ($|m_{H_2}^{net}|$) is calculated by Eq. (43); battery initial energy consumption ($W_{ba,ini}$) is calculated by Eq. (41); system total energy efficiency (f_{sys}) is calculated as Eq. (40)).

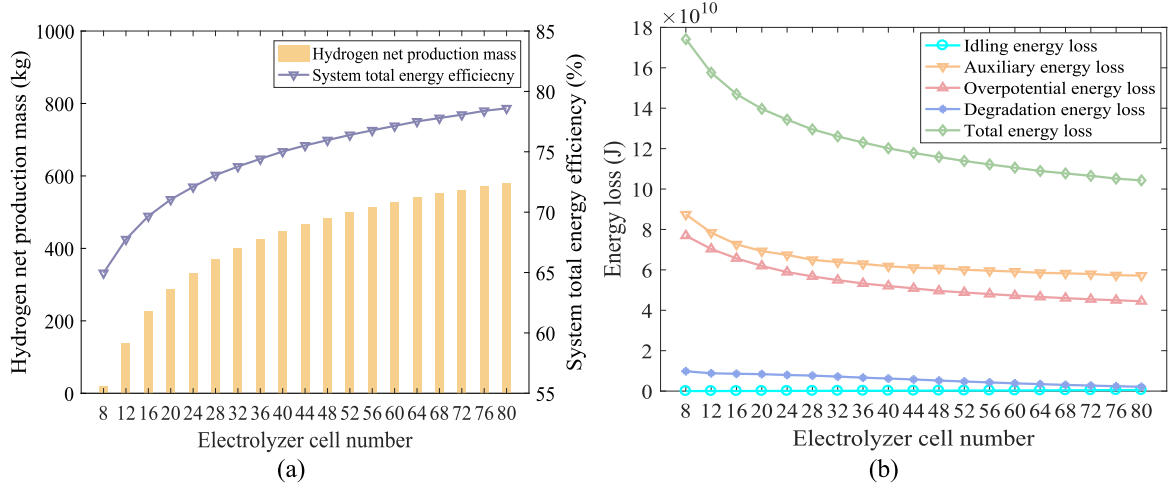


Fig. 8. H₂ net production mass, total energy efficiency of the CCHP system and electrolyzer energy loss in the whole-year operation: (a) H₂ net production mass ($|m_{H_2}^{net}|$) and total energy efficiency (f_{sys}) of the CCHP system; (b) idling energy loss ($W_{el,idling}$), auxiliary energy loss ($W_{el,auxi}$), overpotential energy loss ($W_{el,E}$), degradation energy loss ($W_{el,de}$) and total energy loss ($W_{el,loss}$) of the electrolyzer. (Notes: H₂ net production mass ($|m_{H_2}^{net}|$) is calculated by Eq. (43); total energy efficiency is calculated by Eq. (40); electrolyzer energy loss calculation refers to Eqs. (15)–(18)).

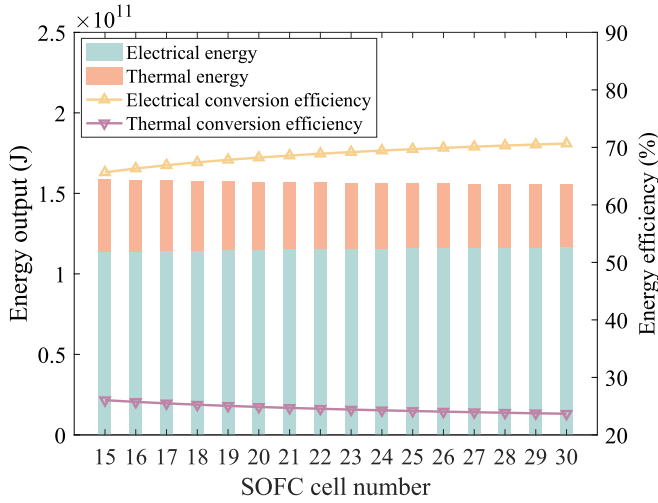


Fig. 9. Impact of SOFC cell number designs on output electrical energy, output thermal energy, electrical energy efficiency and thermal energy efficiency of H₂ energy after the whole-year operation. (Note: electrical energy is output by SOFC stack, and thermal energy is output by both SOFC stack and afterburner with electrochemical reaction and burning).

13.635 %, and the size increase of SOFC from cell number of 15 to 30 will improve the electrical energy efficiency by 4.996 % and total energy efficiency by 2.194 %.

3.3. Economic performance analysis under degradation

In this section, the CCHP system annual costs will be analyzed in the variable sizing of battery, electrolyzer and SOFC with the performance degradation impacts.

3.3.1. Battery economic performance analysis

The battery capacity degradation after one-year operation will decrease in general with the cell number rise, as shown in Fig. 11. Specifically, the battery capacity degradation amplitude will decrease from 20.556 % to 2.762 % (by 17.794 %) in general with the cell number rise from 180 to 960. The underlying mechanism is the less charging-discharging cycle number in larger battery capacity. In detail, the

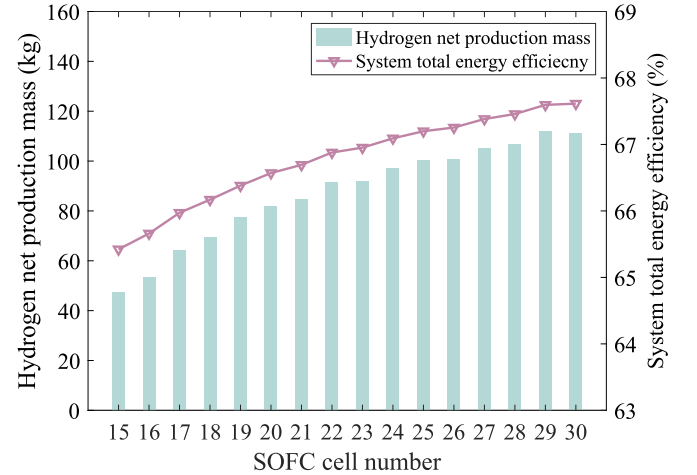


Fig. 10. Hydrogen net production mass and system total energy efficiency in variable SOFC stack cell numbers.

more cell number of the battery, the higher energy capacity, leading to less charging-discharging cycles when covering a similar energy demand. The low frequency of charging-discharging cycle will cause less damage to the battery electrolyte.

Besides, the battery capacity degradation is also dependent on the DOD of each charging-discharging cycle. For example, the battery capacity will decline by 4.186 % in cell number of 540, but by 6.255 % in cell number of 600. The reason is that the battery contains more charging-discharging cycles with high DODs in cell number of 600 than that in cell number of 540, and high DOD will cause large damage to the battery. In detail, battery contains 5 charging-discharging cycles with $DOD \geq 0.5$ in cell number of 600, but the DODs are all below 0.5 in the cell number of 540 design. The different battery DODs of charging-discharging cycles result from the mismatch between power demand output and intermittent solar power input, as described in Fig. 7(b).

Fig. 12 exhibits the variation of annual total cost, initial investment cost, maintenance cost and depreciation cost of the CCHP system along with battery cell number increase. The result indicates that CCHP system annual total cost has little correlations to the battery size. Specifically, the CCHP system annual total cost fluctuates in the range from 1.193×10^5 \$ to 1.299×10^5 \$ without any trend with cell number variation. It is

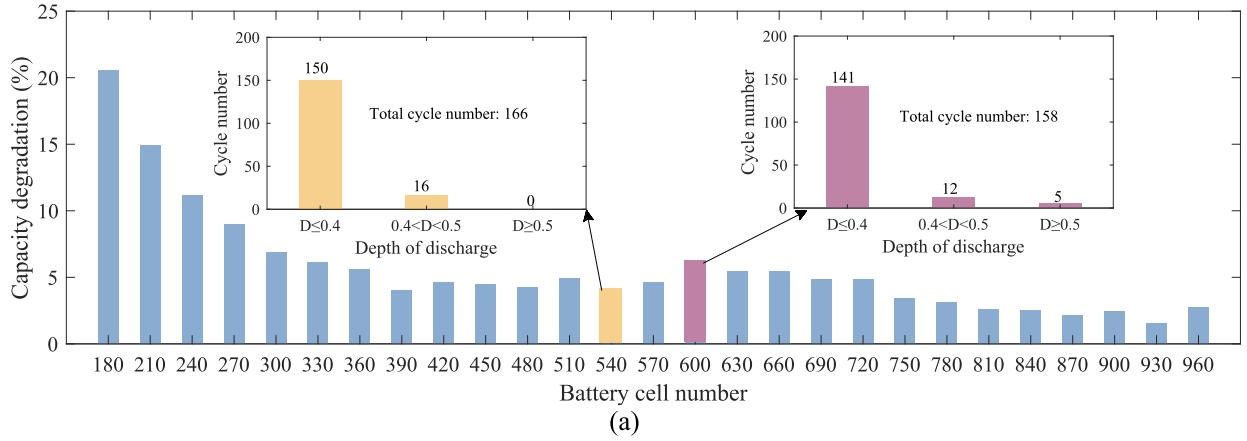


Fig. 11. Battery degradation amplitudes in different cell numbers during the whole-year operation, and battery cycle numbers under different depth of discharge in cell number designs of 540 and 600.

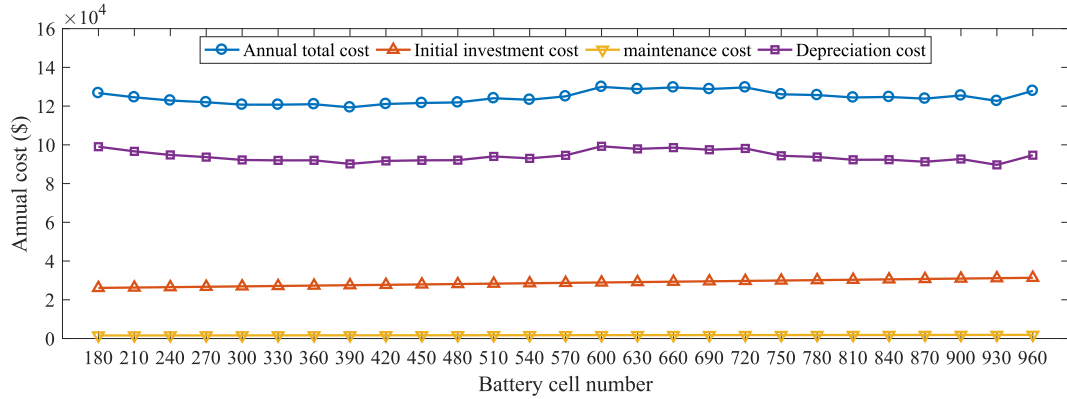


Fig. 12. CCHP system annual total cost, initial investment cost, maintenance cost and depreciation cost with battery variation from 180 to 960.

because of the offset effects between initial investment cost increment and depreciation cost reduction. Specifically, the initial investment cost increases from 2.612×10^4 \$ to 3.136×10^4 \$ with cell number rising

from 180 to 960, but the depreciation cost decreases from 9.904×10^4 \$ to 9.466×10^4 \$ due to the capacity degradation mitigation with cell number rise.

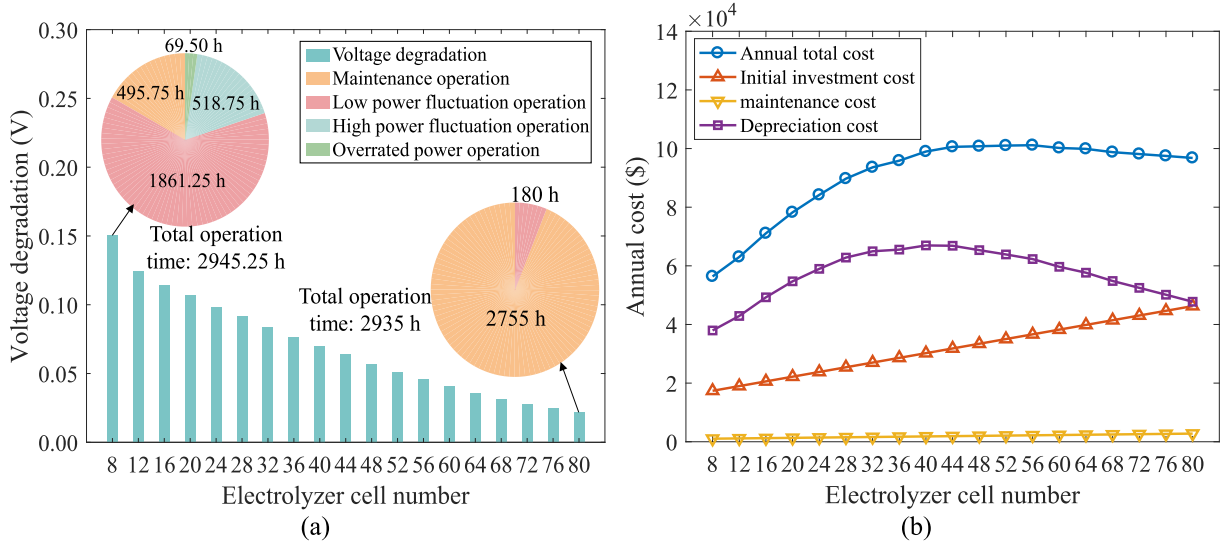


Fig. 13. Electrolyzer degradation voltage and CCHP system annual costs in variable cell numbers: (a) electrolyzer degradation voltage and period-operating time of representative sizing designs; (b) CCHP system annual total cost, initial investment cost, maintenance cost and depreciation cost in variable cell numbers. (Note: Maintenance operation, low power fluctuation operation, high power fluctuation operation and overrated power operation of electrolyzer refer to the input power density of ≤ 0.6 W/cm², $0.6\text{--}4$ W/cm², 4 W/cm², $4\text{--}6$ W/cm², and > 6 W/cm²).

Therefore, the battery capacity degradation is dependent on both size design and charging cycle time, with degradation amplitude declining from 20.556 % to 2.762 % when cell number rising from 180 to 960. Meanwhile, annual total cost of the CCHP system is independent on the battery size due to the offset effects between battery initial investment cost and depreciation cost variation.

3.3.2. Electrolyzer economic performance analysis

The electrolyzer voltage degradation will be relieved with cell number rise after whole-year operation, as shown in Fig. 13(a). In detail, the electrolyzer cell voltage degradation amplitude will decrease from 0.150 V to 0.022 V (by 0.128 V) with the cell number rise from 8 to 80. The reason is the decreasing cell power of electrolyzer with the cell number rise, and lower power held by each cell will cause a slighter damage to the electrolyzer materials. For reference, the electrolyzer cell will hold high power density of 4–6 W/cm² (high power fluctuation period) for 518.75 h and >6 W/cm² (overrated power period) for 69.50 h in cell number of 8, but it will only hold power density of ≤4 W/cm² (maintenance period and low power fluctuation period) in cell number of 80.

The CCHP system annual total cost will be raised in general when the electrolyzer cell number increases, as shown in Fig. 13(b). Specifically, the CCHP system annual total cost will increase from 5.637×10^4 \$ to 9.679×10^4 \$ (by 4.042×10^4 \$) with the electrolyzer cell number rising from 8 to 80. The underlying mechanism is that the increment of initial investment cost (resulting from electrolyzer investment increase) will overwhelm the decrement of depreciation cost (caused by voltage degradation mitigation) with the cell number rise.

Therefore, the capacity enlargement of electrolyzer (cell number rising from 8 to 80) will mitigate its voltage degradation amplitude (from 0.150 V to 0.022 V), but increase the CCHP system annual total cost (from 5.637×10^4 \$ to 9.679×10^4 \$).

3.3.3. SOFC economic performance analysis

The SOFC degradation will be mitigated with its cell number increase, as shown in Fig. 14(a). For example, the SOFC voltage degradation rate will decline from 3.590 % to 1.412 % (by 2.178 %) with the cell number rise from 15 to 30. The reason is that the output power shared by each SOFC cell is reduced with cell number increase, leading to a lower cell voltage (shown in Fig. 2(a)), and lower cell voltage of

SOFC will play a slighter damage on the electrolyte. To be more specific, the SOFC will operate for 273.50 h with cell voltage <0.7 V in cell number of 15, but it will operate with cell voltage >0.7 V all the time in cell number of 30.

The annual total cost of the CCHP system will be increased with the increase in SOFC cell number, as shown in Fig. 14(b). Specifically, the CCHP system annual total cost will increase from 1.200×10^5 \$ to 1.224×10^5 \$ (by 2.389×10^3 \$) when the SOFC cell number rises from 15 to 30. The reason is similar to that of electrolyzer, i.e., along with the SOFC cell number increase, the increment of initial investment cost overwhelms the decrement of depreciation cost due to the SOFC degradation mitigation.

Therefore, the SOFC size enlargement will relieve its performance degradation, with voltage degradation rate declining from 3.590 % to 1.412 % when cell number rising from 15 to 30. Besides, it will also increase the annual total cost of the CCHP system from 1.200×10^5 \$ to 1.224×10^5 \$.

Overall, the battery capacity degradation is dependent on both its size and charging cycles, and its capacity degradation will be mitigated from 20.556 % to 2.762 % with its cell number rising from 180 to 960. The size enlargement of electrolyzer and SOFC will mitigate their performance degradation, but increase the annual cost of the CCHP system. The case study shows the increase in system annual total cost by 4.042×10^4 \$ with the electrolyzer cell number rising from 8 to 80, and by 2.389×10^3 \$ with the SOFC cell number rising from 15 to 30.

3.4. Multi-objective optimization and posteriori multi-criteria decision making for integrated energy system sizing

Based on the above analysis, it is obvious that the size designs of the key devices play trade-off roles in total energy efficiency and annual total cost of the CCHP system. The device size enlargement will improve the system total energy efficiency but increase the annual total cost. In this section, the sizes of battery, electrolyzer, and SOFC will be optimized with objectives of annual total cost and total energy efficiency by calculating 17500 cases based on the NSGA-II algorithm. The contributions of each device size to the system performances are then quantified by ANOVA analysis. Besides, along the Pareto front optimal cases, ‘the best of the best’ cases are then identified by normalized Euclidean distance judging, so as to guide the optimal design and operation.

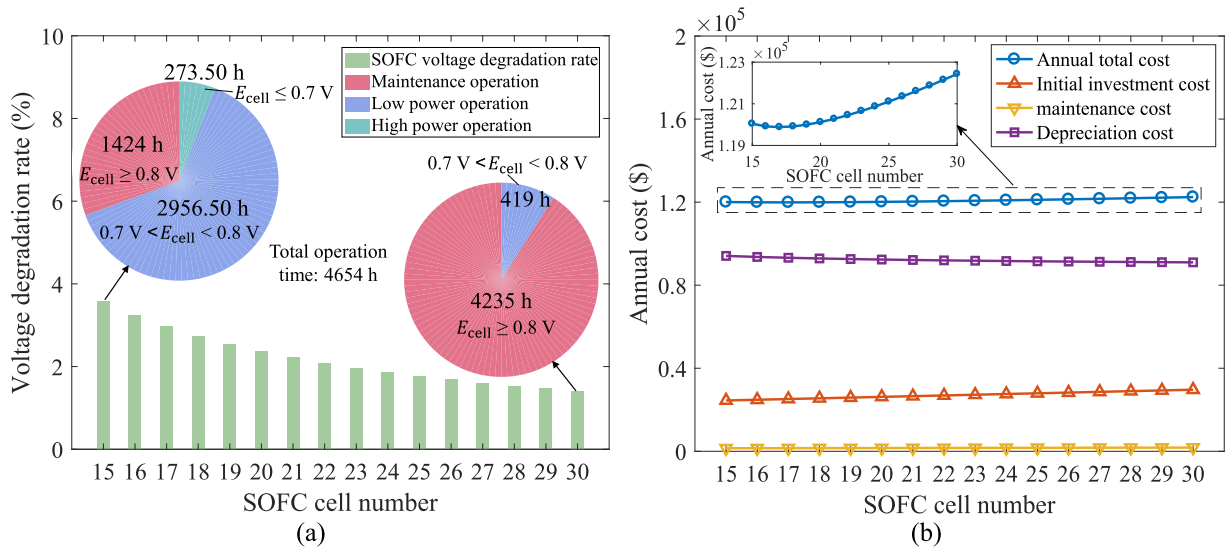


Fig. 14. SOFC voltage degradation rate and CCHP system annual cost in variable cell numbers: (a) SOFC voltage degradation rate and period-operating time of representative cell number designs; (b) CCHP system annual total cost, initial investment cost, maintenance cost, and depreciation cost in variable cell numbers. (Note: maintenance operation, low power operation and high power operation refer to the durations of SOFC cell voltage $E_{cell} \geq 0.8$ V, $0.7 < E_{cell} < 0.8$ V, and $E_{cell} \leq 0.7$ V, respectively.).

3.4.1. Co-effect and contribution analyses

Fig. 15 exhibits the co-effects of device sizes on the annual total cost and total energy efficiency of the CCHP system throughout the multi-objective optimization process with 17500 sizing cases. It is distinct that the battery and SOFC sizes have little impact on system annual total cost compared to that of the electrolyzer, and cases of low annual total cost are mostly centralized in the scope of small electrolyzer size designs, as shown in Fig. 15(a). Specifically, annual total cost of the CCHP system will be lower than 7.220×10^4 \$ when the electrolyzer cell number is less than 30. The reason is that, along with the increase in cell number, the overwhelming effect of initial investment cost increase to the depreciation cost decrease is more obvious in electrolyzer than that of battery and SOFC, due to the slighter degradation mitigation in electrolyzer with cell number rise.

Meanwhile, the cases of high system total energy efficiency are centrally distributed in the scope of large electrolyzer size designs, as shown in Fig. 15(b). For instance, the system total energy efficiency will be above 77.250 % when the electrolyzer cell number is more than 65. The underlying mechanism is that the enlargement of electrolyzer size will reduce the energy loss distinctly (as shown in Fig. 8(b)), and the conversion efficiency from solar energy to H_2 energy will be improved significantly.

The contribution ratios of the battery, electrolyzer and SOFC sizes to the CCHP system annual total cost and total energy efficiency are identified by ANOVA analysis based on the 17500 design cases, and the results are exhibited in Table 5. According to Table 5, the electrolyzer size design plays the most significant role in both annual total cost performance and total energy efficiency performance, occupying the contribution ratios of 0.996 and 0.991, respectively. It indicates that the sizing adjustment of electrolyzer can effectively improve the CCHP system techno-economic performances, which needs to be prioritized compared to other devices. Besides, the contribution ratios of battery (0.007 in total energy efficiency and 0.003 in annual total cost) and SOFC (0.002 in total energy efficiency and 0.001 in annual total cost) are very close but much lower than that of electrolyzer. It indicates that adjusting the battery and SOFC sizes plays slight roles in system performance improvement. Therefore, adopting an electrolyzer with low-cost but high-power capacity in practice is the most effective to improve the overall performances of the CCHP system.

3.4.2. Posteriori multi-criteria decision making from Pareto front

The results of multi-objective optimization for both optimal CCHP system annual total cost and total energy efficiency are shown in Fig. 16, including all the study cases and Pareto front cases. As shown in Fig. 16

(a), by optimally designing device sizes, the annual total cost can be reduced from 1.019×10^5 \$ to 4.856×10^4 \$, and the total energy efficiency can be improved from 64.661 % to 79.997 %. Fig. 16(b) shows the normalized study cases and Pareto cases, with 10 selected ‘best of the best’ optimal cases according to the ascending sort of normalized Euclidean distance relative to the optimal case coordinate (0,1), as exhibited in Table 6. Distinctly, the best sizes of battery, electrolyzer and SOFC are in the cell number range of 333–403, 17–20, and 26–30, respectively. The corresponding optimal range of total energy efficiency is in 70.861 %–72.147 %, and the optimal range of annual total cost is within 6.723×10^4 \$– 7.325×10^4 \$.

Therefore, the electrolyzer size design contributes to the CCHP system annual total cost and total energy efficiency mostly, occupying 0.996 and 0.991 respectively based on the ANOVA analysis. The optimal sizes of battery, electrolyzer and SOFC are in the cell number range of 333–403, 17–20, and 26–30, respectively. The best ranges of total energy efficiency and annual total cost can be controlled within 70.861 %–72.147 % and 6.723×10^4 \$– 7.325×10^4 \$, respectively.

Overall, the study proposed a multi-software optimization platform to analyze the optimal key device sizes of a CCHP system consisting of PV, electrolyzer, battery, SOFC and absorption chiller. The SOFC semi-empirical model is established to substitute the SOFC module for high-efficiency interaction with BOP component models. Impacts of battery, electrolyzer and SOFC sizes on system total energy efficiency, device degradations and annual total cost has been comprehensively investigated. The optimal size designs of the key devices have been calculated by multi-objective optimization, and their contribution ratios to system techno-economic performances have been evaluated by ANOVA analysis. However, some limitations need to be noticed as follows:

- (1) The fitting coefficient estimation of SOFC semi-empirical model needs to be updated when the operating parameters change, and the updated process of the semi-empirical model will delay the operating simulation to wait for the training database generation by the SOFC theoretical model. Further improvement in calculation efficiency is still necessary.
- (2) The SOFC model accuracy is validated in a limited range of operating parameters, such as the operating temperature in 550°C and 600 °C, and the generalization of the model can be further improved.
- (3) The sensitivity and contributions of device sizes are influenced by the cell number selection boundary and operating parameter settings (such as operating temperature, etc). The contribution

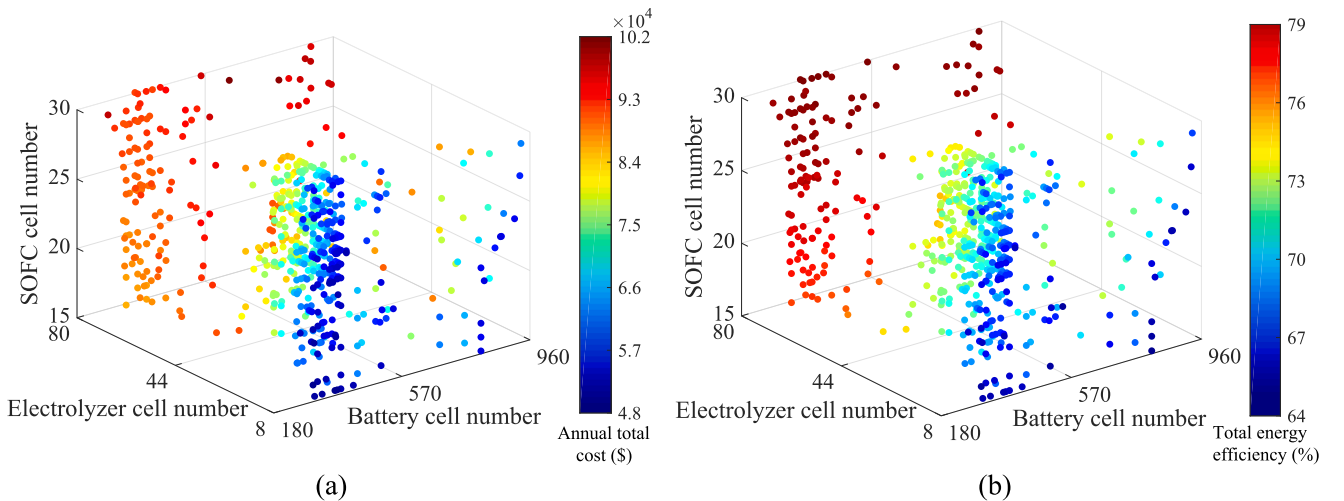


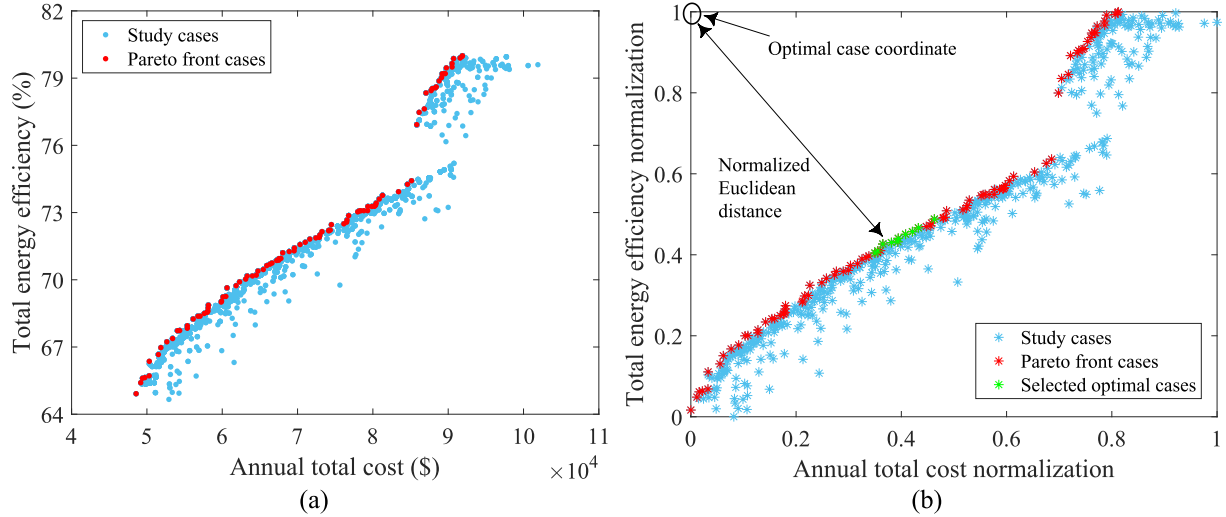
Fig. 15. Multi-objective optimization with impacts of battery, electrolyzer and SOFC cell numbers on the CCHP system annual total cost and total energy efficiency: (a) annual total cost; (b) total energy efficiency.

Table 5

ANOVA analysis results for device sizing contributions to the system performances.

Objectives	Devices	DFF	DFF	MSF	MSE	F	Contribution ratios
Total energy efficiency	Battery	78	17,421	3.003×10^3	7.608	394.716	0.007
	Electrolyzer	46	17,453	1.264×10^4	0.195	6.482×10^4	0.991
	SOFC	15	17,484	2.128×10^3	12.850	165.603	0.002
Annual total cost	Battery	78	17,421	5.052×10^{10}	9.073×10^7	556.817	0.003
	Electrolyzer	46	17,453	1.895×10^{11}	1.041×10^6	1.821×10^5	0.996
	SOFC	15	17,484	3.151×10^{10}	1.394×10^8	226.040	0.001

(Note: The calculation process can refer to the Eqs. (49) – (53)).

**Fig. 16.** Multi-objective optimisation results of the CCHP system annual total cost and total energy efficiency, including all study cases and pareto front cases: (a) study cases before normalization; (b) study cases after normalization.**Table 6**

Selected optimal cases of key device sizing parameters.

Rank	Device sizing parameters			Total energy efficiency (%)	Annual total cost (\$)	Normalized Euclidean distance
	Battery cell number	Electrolyzer cell number	SOFC cell number			
1	394	17	30	71.204	6.803×10^4	0.680
2	403	18	28	71.419	6.959×10^4	0.684
3	403	18	30	71.572	7.033×10^4	0.684
4	370	18	27	71.258	6.889×10^4	0.686
5	370	20	26	71.815	7.163×10^4	0.687
6	360	19	28	71.672	7.102×10^4	0.687
7	378	17	28	70.943	6.752×10^4	0.689
8	360	20	30	72.147	7.325×10^4	0.690
9	333	18	28	71.289	6.955×10^4	0.691
10	403	17	26	70.861	6.723×10^4	0.691

(Note: The calculation process can refer to the Eqs. (54) – (55)).

ratios and optimal designs of the device sizes would be different when the analyzed cell number boundaries and operating parameters are changed.

- (4) Significance of each criterion has not been considered when making decisions to select the optimal cases.

Based on the above limitations, the future work will focus on these aspects:

- (1) A more comprehensive and universal surrogate model of SOFC will be established combining the machine learning techniques and semi-empirical models, aiming to reduce the period of delayed time in variable operating conditions.

- (2) The SOFC model can be further calibrated in a larger range of operating parameter conditions to improve the model generalization performance and practicability.
- (3) The contribution ratios and optimal designs of device sizes will be further investigated in a larger size boundary and wider operating parameter ranges to obtain more comprehensive size design cases, and guide the system size designs in different capacity demands and operating conditions.
- (4) A multi-criteria decision strategy can be developed for the device sizing decision considering the significances of different criteria, in order to satisfy the designed requirements adaptive to the variable application scenarios.

4. Conclusions

In this study, a multi-software optimization platform is established based on MATLAB-TRNSYS co-simulation framework for a CCHP system key device sizing (i.e., battery, electrolyzer and SOFC). A SOFC semi-empirical model is developed and integrated with the co-simulation framework to substitute the SOFC stack for interacting with other BOP models for high-efficiency response. The size effects of battery, electrolyzer and SOFC are analyzed comprehensively on the system total energy efficiency, device degradation level and annual total cost. Besides, the optimal size designs of battery, electrolyzer and SOFC are investigated by multi-objective optimization, considering optimal annual total cost and total energy efficiency. Meanwhile, the contribution ratios of the key device sizes are identified on system techno-economic performances by ANOVA analysis. The main results are summarized as follows:

- 1) To well balance the modelling complexity and computational efficiency CCHP system, the developed SOFC semi-empirical model is highly accurate in both theoretical model surrogate and SOFC performance prediction. The SOFC semi-empirical model can surrogate the theoretical model with fitting R^2 of 1.000 and 0.999 in 550 °C and 600 °C conditions, respectively. Meanwhile, the fitting R^2 between semi-empirical model data and experimental data reach 0.959 and 0.909 in 550 °C and 600 °C conditions, respectively.
- 2) The size enlargement of electrolyzer and SOFC will improve the CCHP system total energy efficiency but increase the annual total cost. The system total energy efficiency will be improved by 13.635 % but annual total cost will be increased by 4.042×10^4 \$, with electrolyzer cell number rising from 8 to 80. The electrical energy efficiency and total energy efficiency will be enhanced by 4.996 % and 2.194 % respectively, but annual total cost will be increased by 2.389×10^3 \$ with SOFC cell number rising from 15 to 30. The electrolyzer size design contributes to the CCHP system techno-economic performances mostly, occupying 0.991 on total energy efficiency and 0.996 on annual total cost, respectively.
- 3) The posteriori multi-criteria decision making based on the multi-objective optimization is effective to obtain the optimal the total energy efficiency and annual total cost of the CCHP system. The 10 advanced optimal size designs of battery, electrolyzer and SOFC are in the cell number range of 333–403, 17–20, and 26–30, respectively, and the corresponding optimal total energy efficiency and annual total cost are within 70.861 %–72.147 %, and 6.723×10^4 \$ – 7.325×10^4 \$, respectively.

The research results provide a new modeling method by integrating with semi-empirical surrogate model to improve the co-simulation efficiency, and give a guidance for the CCHP system sizing for excellent techno-economic feasibility. Besides, the contribution rank analysis figures out the device significances in the CCHP system design. In addition, the research work contributes to improving the widespread application of hydrogen energy in buildings, and accelerates the decarbonization process in communities, which will enable the carbon neutrality in districts for extreme climate mitigation.

CRedit authorship contribution statement

Bin Gao: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yuekuan Zhou:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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